Uniform Bounds on the Influence of Independent and Pairwise Independent Variables

Johannes Gerd Becker*

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Abstract. For independent random variables $X_1, \dots, X_n$, an arbitrary random variable Z, and a suitable class $\mathcal{F}$ of functions, we show that

$$\mathrm{E} \sum_{i=1}^{n} \operatorname*{ess\,sup}_{f \in \mathcal{F}} \big(\mathrm{E}[f(Z) \mid X_i] - \mathrm{E} f(Z) \big)^2 \leq 4 \mathcal{R}_2(\mathcal{F})^2,$$

where $\mathcal{R}_2$ is an ℓ_2-normalized Rademacher complexity. For real-valued Z, the sum of expected squared uniform distances between conditional and unconditional distribution functions is therefore at most 4, independently of n. Under mere pairwise independence, a bracketing argument yields a bound of order $(\log n)^2$. Both orders are sharp up to constants, the latter by an example based on a pairwise-independent random-walk construction of Narayanan. The results also yield population-size-independent bounds on the number of agents who are distributionally pivotal for a collective outcome.

Keywords: Influence bounds; Pairwise independence; Conditional distribution functions; Rademacher complexity; Bracketing; Distributional pivotality; Sensitivity; Large games.

1. Introduction

Let $X_1, \dots, X_n$ be random variables and Z a real-valued random variable on a common probability space. We assume neither that Z is independent of the X_i nor that it is

*ZHAW Zurich University of Applied Sciences, School of Management and Law, Institute for Risk & Insurance, Winterthur, Switzerland. johannes.becker@zhaw.ch

a function of them. In this note, we ask how strongly a single X_i can influence the *distribution* of Z, measured by the uniform distance

$$D_i := \sup_{z\in\mathbb{R}} \bigl|F_Z(z \mid X_i) - F_Z(z)\bigr|$$

between the conditional and the unconditional distribution function.

For a single functional of Z, the answer is classical. If the X_i are pairwise independent and $f(Z) \in L^2$, the centred conditional expectations $\mathrm{E}[f(Z) \mid X_i] - \mathrm{E}f(Z)$ are mutually orthogonal, and Bessel's inequality gives

$$\sum_{i=1}^{n} \operatorname{Var} \mathrm{E}\bigl[f(Z) \;\big|\; X_i\bigr] \le \operatorname{Var} f(Z) \tag{1}$$

(Lemma 18). For $f = \mathbf{1}_{(-\infty,z]}$, this yields $\sum_i \mathrm{E}\bigl(F_Z(z \mid X_i) - F_Z(z)\bigr)^2 \le 1/4$ for every fixed z. The question is what happens when the supremum over z is taken inside the sum.

The answer depends on the dependence structure of the X_i in a sharp way. If the X_i are independent, then

$$\mathrm{E}\sum_{i=1}^{n} D_i^2 \le 4 \tag{2}$$

for every n and every law of Z (Corollary 9). This is a special case of a bound for general classes $\mathcal{F}$ of functionals in terms of an ℓ_2-normalized Rademacher complexity $\mathcal{R}_2(\mathcal{F})$ (Theorem 7). The n-independent order is optimal: an example shows that $\mathrm{E}\sum_i D_i^2$ exceeds $1/4$ for $n \ge 6$ and tends to at least $2/\pi$. If the X_i are only pairwise independent, (2) fails. We show that

$$\mathrm{E}\sum_{i=1}^{n} D_i^2 \le \left(1 + \left\lceil \frac{1}{4}\log_2 n \right\rceil\right)^2$$

(Corollary 20), and an example based on a pairwise independent random walk constructed by Narayanan [19] shows that the order $(\log n)^2$ is attained (Section 4). Pairwise independence thus gives an n-independent bound for every single functional of Z, but not for its conditional distribution function as a whole.

Methods. The methods we use are borrowed from empirical process theory [29], albeit applied to a somewhat uncommon functional, namely $E\bigl[f(Z) \;\big|\; X_i\bigr] - \mathrm{E}f(Z)$. For independent X_i, we symmetrize and bound a Rademacher average; for indicators of half-lines, Lévy's maximal inequality gives $\mathcal{R}_2 \le 1$. For pairwise independent X_i, the symmetrization argument breaks down, and we chain (1) over subsequently finer grids approximating a bracket cover instead.

AI systems, in particular GPT-6 Astra and, to a lesser extent, Claude Opus 5.5, contributed significantly to the results presented here. We shall give a detailed description at the end of the paper, before the References section.

Economic motivation. Our problem originates in the theory of large games and mechanisms, where X_i is the private type of agent i and Z a collective outcome, such as the provision of a public good or a vote share. Al-Najjar and Smorodinsky [17], motivated by the public-goods problem of Mailath and Postlewaite [15, p. 364], call an agent α-*pivotal* if the expected outcome conditional on his type varies by at least α across types, and gave tight bounds on the number of α-pivotal agents for bounded outcomes and independent finite-valued types with type probabilities uniformly bounded away from zero. Gradwohl et al. [11] introduced (p, α)-*pivotality*, under which the conditional expected outcome deviates from the unconditional one by more than α with a probability of more than p, and extended the bounds to pairwise independent types. Both papers argue combinatorially. Becker [1, Prop. 2.1] observed that inequality (1)—an orthogonality argument used in Appendix 1 of Mailath and Postlewaite [15]—gives such bounds directly: there are at most $\operatorname{Var} f(Z)/(p\alpha^2)$ agents that are (p, α)-pivotal for $f(Z)$. In economic terms, the variances of the type-conditional expected outcomes add up to at most the unconditional variance of the common outcome. The argument does not require finite type spaces.

The present paper extends this result from a single functional to the whole distribution of the outcome. Call agent i *distributionally* (p, α)-*pivotal* if $D_i \geq \alpha$ with probability at least p. By inequality (2), which we show in the present paper, there are at most $4/(p\alpha^2)$ such agents when types are independent, irrespective of the number of agents and of the outcome. Under pairwise independence, we obtain a bound that is uniform in the number of agents as well, yet $\mathrm{O}\big((\log(1/\alpha))^2/(p\alpha^2)\big)$ for $\alpha \downarrow 0$ or $p \downarrow 0$ (Section 5.1).

For clarity, however, we should note that without additional assumptions, this pivotality is an actual, not a counterfactual, property. In a strategic game, it only describes the outcome as the agents actually behave, that is, along the equilibrium path. It does not, in general, say which agents, or how many of them, *could* shift the outcome if they acted differently. If, for instance, all players vote in favour under a unanimity rule, no player is pivotal as just defined, yet every player could change the outcome by voting against.

In some games, however, the two notions coincide. In an *incentive-compatible direct mechanism*, each agent reports a type, the outcome is computed from the reports, and truthful reporting is an equilibrium [15, 17]. If types are independent and all types have strictly positive probability, an agent who reports x' instead of his true type produces the outcome distribution $F_Z(\cdot \mid X_i = x')$. The distance from the truthful outcome distribution at type x to the distribution induced by report x' satisfies

$$\sup_z \big| F_Z(z \mid X_i = x') - F_Z(z \mid X_i = x) \big| \leq D_i(x') + D_i(x),$$

so the term on the right-hand side bounds how far agent i can shift the conditional distribution of the outcome by misreporting. Note, however, that under pairwise independence, this argument would already no longer hold, and our example would concern the defined concept of pivotality only.

When agents observe other's actions only with noise, pivotality bounds become strategically relevant as well. In a game with one large player and many small players, Fudenberg, Levine, and Pesendorfer [9] show, under their noise and payoff assumptions, and with the number of the large player's actions bounded independently of the number of small players, that the equilibrium set converges to that of the corresponding game with a continuum of small players. As the population grows, all but a vanishing fraction of small players behave as if they were negligible. Al-Najjar and Smorodinsky [18] subsequently derive, under suitable assumptions, that at each history all but a number of players bounded independently of population play approximate one-shot best responses.

Sensitivity analysis. The distance between conditional and unconditional distribution functions is also employed in sensitivity analysis of numerical models [23]. Suppose $X_1, \dots, X_n$ are uncertain inputs and $Z = g(X_1, \dots, X_n)$ is a model output. The inputs may, for instance, be model parameters, initial or boundary conditions, or forcing data. As Pianosi et al. [22] explain in their review, global sensitivity analysis asks how much of the variation in the output is due to each input, considering the entire feasible range of the input, in order to identify influential inputs, prioritize efforts to reduce uncertainty (for instance, by guiding further measurements), or understand model behaviour.

A classical approach measures sensitivity through variance [27, 26]. For a square-integrable output with positive variance, the first-order Sobol' index of input X_i is [26, Chapter 4, p. 161, Eq. (4.9)]

$$S_i = \frac{\operatorname{Var}(\mathrm{E}[Z \mid X_i])}{\operatorname{Var} Z}\,.$$

Under independence of the inputs, these indices sum to at most one, which also follows directly from (1). As these first-order indices capture only variation in the conditional mean, they cannot detect an input that changes the spread or shape of the output distribution while leaving the conditional mean unchanged; distribution-based measures can capture such effects [5, 23].

The PAWN method of Pianosi and Wagener [23] uses the Kolmogorov–Smirnov distance between the conditional and unconditional output distributions. Its basic quantity is

$$D_i(x_i) := \sup_{z \in \mathbb{R}} \big| F_Z(z \mid X_i = x_i) - F_Z(z) \big|,$$

which PAWN summarizes over conditioning values, for example by the median or maximum. As an application, Pianosi and Wagener study a rainfall-runoff model whose uncertain parameters govern soil moisture storage and the routing of water through different flow pathways, and determine which of them most affect the distribution of the maximum predicted river flow—including sensitivity restricted to a high-flow subrange—and hence which sources of uncertainty deserve closer attention when the model is used to assess flood risk.

Besides $D_i(X_i)$, its expectation—the expected Kolmogorov–Smirnov distance—

$$\beta_i^{\mathrm{KS}} := \mathrm{E} D_i(X_i)$$

is used as a sensitivity index as well [24, Eq. (8), p. 2643]. Our results place a joint restriction on these individual distributional sensitivities. For independent inputs, Jensen's inequality and Corollary 9 give

$$\sum_{i=1}^{n} (\beta_i^{\mathrm{KS}})^2 \le \sum_{i=1}^{n} \mathrm{E} D_i(X_i)^2 \le 4.$$

Consequently, for any $\alpha > 0$, at most $4/\alpha^2$ inputs can have $\beta_i^{\mathrm{KS}} \ge \alpha$, irrespective of the total number of inputs. For pairwise independent inputs, Corollary 20 gives only

$$\sum_{i=1}^{n} (\beta_i^{\mathrm{KS}})^2 \le \sum_{i=1}^{n} \mathrm{E} D_i(X_i)^2 \le \left\lceil 1 + \frac{1}{4} \log_2 n \right\rceil^2 .$$

Rather than considering the supremum, the first-order Cramér–von Mises sensitivity indices of Gamboa, Klein, and Lagnoux [10] *average* squared differences between conditional and unconditional output cumulative distribution functions over the conditioning input, integrate over thresholds with respect to the output distribution $\mathrm{d}F_Z(t)$, and normalize by $\int F_Z(t)(1 - F_Z(t))\,\mathrm{d}F_Z(t)$:

$$S_i^{\mathrm{CvM}} = \frac{\displaystyle\int \mathrm{E}\left[\big(F_Z(t \mid X_i) - F_Z(t)\big)^2\right] \mathrm{d}F_Z(t)}{\displaystyle\int F_Z(t)\big(1 - F_Z(t)\big)\,\mathrm{d}F_Z(t)} .$$

For these indices, already our inequality (1) implies that their sum is at most one even under pairwise independence of the inputs, provided the normalizing denominator is positive. Indeed, applying that inequality to $\mathbf{1}\{Z \le t\}$, integrating with respect to $\mathrm{d}F_Z(t)$, and dividing by the common denominator gives the claimed bound. For the Kolmogorov–Smirnov distance, by contrast, each input's contribution involves its own supremum over thresholds, so the same integration argument does not apply; controlling these suprema is the additional difficulty addressed by our results.

As with the economic interpretation, these sensitivity measures must be interpreted with some care. For example, they describe what observing a *single* input reveals about the output under the chosen joint distribution of the inputs. Small individual sensitivities do not rule out that a group of inputs is *jointly* highly informative: if X_1 and X_2 are independent fair bits and $Z = X_1 \operatorname{xor} X_2$, then $D_1(X_1) = D_2(X_2) = 0$, although (X_1, X_2) determines Z.

2. The independent case

2.1. Complexity measure: Rademacher complexity

Definition 1 (Rademacher complexity)**.** *Let S be a nonempty set and let $\mathcal{F}$ be a set of functions $f\colon S \to \mathbb{R}$. For finite, nonempty $\mathcal{F}$, we define the* ℓ_2-normalized Rademacher

complexity *of* $\mathcal{F}$ *by*

$$\mathcal{R}_2(\mathcal{F}) := \sup\left\{ \mathrm{E}_\epsilon \sup_{s\in S} \sum_{f\in\mathcal{F}} \epsilon_f a_f f(s) \;:\; a \in \mathbb{R}^{\mathcal{F}},\ \|a\|_2 = 1 \right\}, \tag{3}$$

where $(\epsilon_f)_{f\in\mathcal{F}}$ *is a family of independent Rademacher signs, i.e., independent random variables taking the values* $+1$ *and* -1*, each with probability* $1/2$*. We set* $\mathcal{R}_2(\emptyset) := 0$*. For general* $\mathcal{F}$*, we define*

$$\mathcal{R}_2(\mathcal{F}) := \sup\big\{\mathcal{R}_2(\mathcal{T}) : \mathcal{T} \subseteq \mathcal{F} \textit{ finite}\big\} \in [0,\infty]. \tag{4}$$

It is easy to see that for finite sets $\mathcal{F}$, (4) is consistent with (3), as the Rademacher complexity defined in (3) cannot increase when passing to a subset: if $\mathcal{T} \subseteq \mathcal{F}$, any coefficient vector on $\mathcal{T}$ can be extended to $\mathcal{F}$ by setting the remaining coefficients equal to zero. The next lemma shows that $\mathcal{R}_2(\mathcal{F})$ controls arbitrary weighted Rademacher sums of functions from $\mathcal{F}$, allowing in particular for repetitions.

Lemma 2 (Rademacher bound for finite families)**.** *For arbitrary* $g_1, \ldots, g_m \in \mathcal{F}$ *and* $c_1, \ldots, c_m \in \mathbb{R}$*,*

$$\mathrm{E}_\epsilon \sup_{s\in S} \sum_{j=1}^m \epsilon_j c_j g_j(s) \le \mathcal{R}_2(\mathcal{F}) \cdot \left(\sum_{j=1}^m c_j^2\right)^{1/2}.$$

A proof of the lemma is given in the appendix.

Lemma 3 (Invariance under constant shifts)**.** *Let* $\mathcal{F}$ *and* $\mathcal{H}$ *be sets of real-valued functions on the same domain* S*. Suppose that every function in either set differs by a constant from some function in the other set. Then,* $\mathcal{R}_2(\mathcal{F}) = \mathcal{R}_2(\mathcal{H})$*.*

A proof is given in the appendix.

For indicator functions of half-lines, we have the following bound:

Lemma 4 (Rademacher complexity of half-lines)**.** [1] *For the set*

$$\mathcal{I} := \big\{\mathbf{1}_{(-\infty,t]} : t \in \mathbb{R}\big\}$$

of indicator functions of half-lines of the real line, we have

$$\mathcal{R}_2(\mathcal{I}) \le 1.$$

A proof is given in the appendix.

[1] The result can be found in [21, Eq. (3.11)]. The exact optimal constant was left open there, and we are unaware of its having been determined subsequently.

2.2. Influence result for the independent case

To avoid measurability issues, we impose a mild separability condition on the function classes under consideration.

Definition 5 (Admissible set of functions)**.** *Let Z be a random variable with values in a measurable space $(S, \mathcal{S})$, and let P_Z be the distribution of Z. A set $\mathcal{F}$ of functions $f\colon S \to \mathbb{R}$ is called* Z-admissible *if every $f \in \mathcal{F}$ is $\mathcal{S}$-measurable and square-integrable with respect to P_Z, and there exists a countable subset $\mathcal{F}_0 \subseteq \mathcal{F}$ that is dense in $\mathcal{F}$ with respect to the $L^2(P_Z)$-seminorm.*

For completeness, we recall the notion of an essential supremum.

Definition 6 (Essential supremum)**.** *For a family $\big(Y_\alpha\big)_{\alpha\in A}$ of extended-real-valued random variables, the* essential supremum

$$\operatorname*{ess\,sup}_{\alpha\in A} Y_\alpha$$

is the a. s.-equivalence class of the smallest random variable Y such that

$$Y_\alpha \le Y \quad \text{a. s. for every } \alpha \in A.$$

We note that the essential supremum always exists. If the set is finite or countably infinite, it is represented, up to a. s.-equivalence, by the pointwise supremum.

We can now state our first main theorem.

Theorem 7 (Independent case)**.** *Let $X_1, \dots, X_n$ be independent random variables, let Z be a random variable, and let $\mathcal{F}$ be a Z-admissible set. Then*

$$\mathrm{E}\sum_i \operatorname*{ess\,sup}_{f\in\mathcal{F}} \Big(\mathrm{E}\big[f(Z) \mid X_i\big] - \mathrm{E}f(Z)\Big)^2 \le 4\mathcal{R}_2(\mathcal{F})^2. \tag{5}$$

Proof. Reduction to Z-centred families. By Lemma 3,

$$\mathcal{R}_2\big(\{f - \mathrm{E}f(Z) : f \in \mathcal{F}\}\big) = \mathcal{R}_2(\mathcal{F}).$$

This replacement also preserves Z-admissibility, since

$$\big\|\big(f - \mathrm{E}f(Z)\big) - \big(g - \mathrm{E}g(Z)\big)\big\|^2_{L^2(P_Z)} = \mathrm{Var}\big((f-g)(Z)\big) \le \|f - g\|^2_{L^2(P_Z)}$$

and thus the image of a countable dense subset under centring is again countable and dense. Since centring also leaves the left-hand side of (5) unchanged, it suffices to prove the theorem under the assumption that $\mathrm{E}f(Z) = 0$ for every $f \in \mathcal{F}$.

Reduction to the finite case. Assuming that $\mathcal{F}$ is Z-centred, we now reduce the proof to the case where $\mathcal{F}$ is finite. Let $\mathcal{F}_0 = \{f_1, f_2, \ldots\} \subseteq \mathcal{F}$ be dense in $\mathcal{F}$ with respect to the $L^2(P_Z)$-seminorm, and set $\mathcal{F}_m := \{f_1, \ldots, f_m\}$. For $f \in \mathcal{F}$, write

$$d_f(X_i) := \mathrm{E}\big[f(Z) \;\big|\; X_i\big],$$

and define

$$S_i := \sup_{f \in \mathcal{F}_0} d_f(X_i)^2 = \sup_{m \geq 1} \max_{f \in \mathcal{F}_m} d_f(X_i)^2.$$

For each $f \in \mathcal{F}$, choose $(g_k) \subseteq \mathcal{F}_0$ with $g_k \to f$ in $L^2(P_Z)$. By conditional Jensen's inequality, $d_{g_k}(X_i) \to d_f(X_i)$ in $L^2(P)$, and hence almost surely along a subsequence. Thus, for each $f \in \mathcal{F}$, we have $d_f(X_i)^2 \leq S_i$ almost surely, so

$$\operatorname*{ess\,sup}_{f \in \mathcal{F}} d_f(X_i)^2 \leq S_i \quad \text{a. s.}$$

The reverse inequality follows from the inclusion $\mathcal{F}_0 \subseteq \mathcal{F}$ and the countability of $\mathcal{F}_0$. Consequently,

$$\operatorname*{ess\,sup}_{f \in \mathcal{F}} d_f(X_i)^2 = \sup_{m \geq 1} \max_{f \in \mathcal{F}_m} d_f(X_i)^2 \quad \text{a. s.}$$

Since the sets $\mathcal{F}_m$ are increasing, monotone convergence yields

$$\mathrm{E} \sum_i \operatorname*{ess\,sup}_{f \in \mathcal{F}} d_f(X_i)^2 = \lim_{m \to \infty} \mathrm{E} \sum_i \max_{f \in \mathcal{F}_m} d_f(X_i)^2.$$

Thus, once the theorem has been proved for finite families, we can conclude that

$$\mathrm{E} \sum_i \operatorname*{ess\,sup}_{f \in \mathcal{F}} d_f(X_i)^2 \leq 4 \sup_{m \geq 1} \mathcal{R}_2(\mathcal{F}_m)^2 \leq 4 \mathcal{R}_2(\mathcal{F})^2.$$

Finite case. We now assume that $\mathcal{F}$ is Z-centred and finite.

To absorb the index i into the conditioning value, we may assume without loss of generality that the X_i take values in pairwise disjoint measurable spaces, replacing X_i by (i, X_i) if necessary. Let $(E_i, \mathcal{E}_i)$ denote these value spaces, and equip $E := \bigsqcup_i E_i$ with the sigma-algebra $\mathcal{E} := \big\{A \subseteq E : A \cap E_i \in \mathcal{E}_i \text{ for every } i\big\}$. For $f \in \mathcal{F}$, choose an $\mathcal{E}$-measurable function $x \mapsto P[x]f$ such that

$$P[X_i]f = \mathrm{E}\big[f(Z) \;\big|\; X_i\big] \quad \text{a. s. for every } i.$$

Such a choice is possible by the Doob–Dynkin factorization lemma, applied on each E_i to the real-valued, $\sigma(X_i)$-measurable random variable $\mathrm{E}[f(Z) \mid X_i]$, and by combining the resulting functions on the disjoint union E.

Let

$$d_f(x) := P[x]f.$$

For $x \in E$, let $f^*[x]$ be a maximizer of $f \mapsto \big|d_f(x)\big|$, i. e.,

$$f^*[x] \in \operatorname*{argmax}_{f \in \mathcal{F}} \big|d_f(x)\big|,$$

where ties are resolved according to an arbitrary, fixed ordering of $\mathcal{F}$. Set

$$d^*(x) := d_{f^*[x]}(x).$$

Then

$$\big|d^*(x)\big| = \max_{f \in \mathcal{F}} \big|d_f(x)\big|,$$

and the inequality to be proved becomes

$$\mathrm{E} \sum_i \big(d^*(X_i)\big)^2 \le 4\mathcal{R}_2(\mathcal{F})^2.$$

Finally, define

$$f^{**}[x] := d^*(x) f^*[x].$$

First, conditional Jensen's inequality and the square-integrability of the functions in the finite class $\mathcal{F}$ give

$$\mathrm{E}|d^*(X_i)|^2 \le \sum_{f \in \mathcal{F}} \mathrm{E}\big|\mathrm{E}[f(Z) \mid X_i]\big|^2 \le \sum_{f \in \mathcal{F}} \mathrm{E}|f(Z)|^2 < \infty. \tag{6}$$

Since the events $\{f^*[X_i] = f\}$ are $\sigma(X_i)$-measurable, we also have

$$\begin{aligned} \mathrm{E}\big[f^*[X_i](Z) \;\big|\; X_i\big] &= \sum_{f \in \mathcal{F}} \mathbf{1}_{\{f^*[X_i]=f\}} \mathrm{E}[f(Z) \mid X_i] \\ &= \sum_{f \in \mathcal{F}} \mathbf{1}_{\{f^*[X_i]=f\}} P[X_i] f = P[X_i] f^*[X_i]. \end{aligned} \tag{7}$$

Hence, by the definitions of d^* and f^{**},

$$\begin{aligned} \mathrm{E}\big(d^*(X_i)\big)^2 &= \mathrm{E}\Big(d^*(X_i) \cdot P[X_i] f^*[X_i]\Big) \\ &= \mathrm{E}\Big(d^*(X_i) \cdot \mathrm{E}\big[f^*[X_i](Z) \;\big|\; X_i\big]\Big) \\ &= \mathrm{E}\big(f^{**}[X_i](Z)\big). \end{aligned}$$

Introduce a copy $(X_1', \ldots, X_n')$ of $(X_1, \ldots, X_n)$ that is independent of $(X_1, \ldots, X_n, Z)$. Since every $f \in \mathcal{F}$ is centred, we have, in analogy to (7),

$$\begin{aligned} \mathrm{E}\big[f^{**}[X_i'](Z) \;\big|\; X_i'\big] &= d^*(X_i') \cdot \mathrm{E}\big[P f^*[X_i'](Z) \;\big|\; X_i'\big] \\ &= d^*(X_i') \cdot \sum_{f \in \mathcal{F}} \mathbf{1}_{f^*[X_i']=f} \mathrm{E}\big[f(Z) \;\big|\; X_i'\big] \\ &= d^*(X_i') \cdot \sum_{f \in \mathcal{F}} \mathbf{1}_{f^*[X_i']=f} \mathrm{E} f(Z) \\ &= 0. \end{aligned}$$

Consequently,

$$\mathrm{E}\sum_i \big(d^*(X_i)\big)^2 = \mathrm{E}\sum_i f^{**}[X_i](Z) = \mathrm{E}\sum_i \Big(f^{**}[X_i](Z) - f^{**}[X_i'](Z)\Big)$$
$$\leq \mathrm{E}\sup_{s\in S}\sum_i \Big(f^{**}[X_i](s) - f^{**}[X_i'](s)\Big).$$

Introducing Rademacher signs ϵ_i, mutually independent and independent from everything else, this can, by the assumed independence of the X_i and the symmetry of each pair (X_i, X_i'), be rewritten into

$$\mathrm{E}\sup_{s\in S}\sum_i \epsilon_i\Big(f^{**}[X_i](s) - f^{**}[X_i'](s)\Big) = \mathrm{E}\sup_{s\in S}\sum_i \Big(\epsilon_i f^{**}[X_i](s) + (-\epsilon_i) f^{**}[X_i'](s)\Big)$$
$$\leq 2\mathrm{E}\sup_{s\in S}\sum_i \epsilon_i f^{**}[X_i](s),$$

with the inequality following from subadditivity of the supremum and the fact that the two resulting terms have the same distribution. Taking everything together, we have thus reached the inequality

$$\mathrm{E}\sum_i \big(d^*(X_i)\big)^2 \leq 2\mathrm{E}\sup_{s\in S}\sum_i \epsilon_i d^*(X_i) f^*[X_i](s).$$

Conditionally on $X_1, \ldots, X_n$, Lemma 2 gives,

$$\mathrm{E}_\epsilon \sup_{s\in S}\sum_i \epsilon_i d^*(X_i) f^*[X_i](s) \leq \mathcal{R}_2(\mathcal{F}) \cdot \left(\sum_i d^*(X_i)^2\right)^{1/2},$$

Hence, by Jensen's inequality,

$$\mathrm{E}\sum_i \big(d^*(X_i)\big)^2 \leq 2 \cdot \mathcal{R}_2(\mathcal{F}) \cdot \left(\mathrm{E}\sum_i \big(d^*(X_i)\big)^2\right)^{1/2}.$$

Since $\mathrm{E}\sum_i \big(d^*(X_i)\big)^2 < \infty$ by (6), the desired inequality follows. □

Definition 8 (Conditional distribution function)**.** *Let X and Z be random variables, with Z real-valued. We write*

$$F_Z(z) := P(Z \leq z) = \mathrm{E}\mathbf{1}_{\{Z\leq z\}}$$

for the cumulative distribution function of Z, and

$$F_Z(z \mid X) := P(Z \leq z \mid X) = \mathrm{E}\big[\mathbf{1}_{\{Z\leq z\}} \;\big|\; X\big] \quad a.\,s.$$

for a version of the conditional distribution function chosen so that $z \mapsto F_Z(z \mid X)$ is a cumulative distribution function almost surely.

Such a version can be constructed as follows. For each $q \in \mathbb{Q}$, choose a version H_q of $\mathrm{E}[\mathbf{1}_{\{Z \le q\}} \mid X]$. Outside a common null set, $q \mapsto H_q$ is nondecreasing and has limits 0 and 1 at $-\infty$ and $+\infty$, respectively. Outside the exceptional null set, set

$$F_Z(z \mid X) := \inf_{\substack{q \in \mathbb{Q} \\ q > z}} H_q,$$

and on the null set, choose any fixed cumulative distribution function. This construction gives sample paths that are cumulative distribution functions, and conditional monotone convergence yields the required conditional-expectation identity for each $z \in \mathbb{R}$.

Corollary 9 (Cumulative distribution functions, independent case)**.** *Let $X_1, \dots, X_n$ be independent random variables and Z a real-valued random variable. Then*

$$\mathrm{E} \sum_i \sup_{z \in \mathbb{R}} \bigl(F_Z(z \mid X_i) - F_Z(z)\bigr)^2 \le 4.$$

Proof. The family $\mathbf{1}_{(-\infty,t]}$, $t \in \mathbb{Q}$, is countable and consists of bounded measurable functions, so it is Z-admissible, and its essential supremum agrees almost surely with its pointwise supremum. By right-continuity of the conditional and unconditional cumulative distribution functions, this supremum equals the supremum over all real thresholds. The result therefore follows from Theorem 7 and Lemma 4. □

Corollary 10 (Cumulative distribution functions, symmetric independent case)**.** *Let $X_1, \dots, X_n$ be independent random variables and Z a real-valued random variable. Suppose that the distribution of $(Z, X_1, \dots, X_n)$ is invariant under permutations of the X_i. Then, for every i,*

$$\mathrm{E} \sup_{z \in \mathbb{R}} \bigl(F_Z(z \mid X_i) - F_Z(z)\bigr)^2 \le \frac{4}{n}.$$

Proof. By symmetry, all summands in Corollary 9 have the same expectation, so the claim follows upon dividing its bound by n. □

3. The pairwise independent case

3.1. Complexity measure: Variance bracketing complexity

Definition 11 (Bracket, bracket cover)**.** *Let Z be a random variable. For $\ell, u \in L^2(P_Z)$ with $\ell \le u$ P_Z-a. e., the* bracket $[\ell, u]$ *is the set*

$$[\ell, u] := \{f \in L^2(P_Z) : \ell \le f \le u \ P_Z\text{-a.e.}\}.$$

A bracket cover *of $\mathcal{F} \subseteq L^2(P_Z)$ is a family of brackets whose union contains $\mathcal{F}$.*

Definition 12 (Variance pseudometric)**.** *Let Z be a random variable. Write the* variance pseudometric *on $L^2(P_Z)$ as*

$$\sigma_Z(f,g) := \sqrt{\operatorname{Var}\big((f-g)(Z)\big)}$$

and define

$$\sigma_Z(f,\mathcal{W}) := \min_{w\in\mathcal{W}} \sigma_Z(f,w)$$

for nonempty finite $\mathcal{W} \subseteq L^2(P_Z)$.

Definition 13 (Variance bracketing complexity)**.** *Let Z be a random variable, $\mathcal{F} \subseteq L^2(P_Z)$, and $\delta > 0$.*

Consider triples $\big(K, (\mathcal{W}_k)_{k=0}^K, \mathcal{C}\big)$, where $K \in \mathbb{N}_0$, $\mathcal{W}_0, \ldots, \mathcal{W}_K \subseteq L^2(P_Z)$ are nonempty finite sets with $\mathcal{W}_0$ containing only a single element $\{g_0\}$, and $\mathcal{C}$ is a nonempty finite bracket cover of $\mathcal{F}$ with bracket endpoints in $L^2(P_Z)$ such that

$$\max_{[\ell,u]\in\mathcal{C}} \mathrm{E}\big[(u-\ell)(Z)\big] < \delta.$$

The variance bracketing complexity of $\mathcal{F}$ at resolution δ *is then defined to be*

$$\mathcal{V}_Z(\delta,\mathcal{F}) := \inf\Bigg\{ \sqrt{\operatorname{Var} g_0(Z)} + \sum_{k=1}^{K} \sqrt{\sum_{w\in\mathcal{W}_k} \sigma_Z^2(w,\mathcal{W}_{k-1})} + \sqrt{\sum_{[\ell,u]\in\mathcal{C}} \Big(\sigma_Z^2(\ell,\mathcal{W}_K) + \sigma_Z^2(u,\mathcal{W}_K)\Big)} \Bigg\},$$

where the infimum is over all triples considered. Empty sums are zero, and $\inf\emptyset := \infty$.

Since the term has been designed to fit exactly what we shall need for the below Theorem 19, we afford a little detour, which is not required for reaching the theorem. The main path of exposition is to be continued with Lemma 16.

The first part of our detour is the observation that variance bracketing complexity is closely related to L^2-bracketing numbers of decreasing magnitude. Indeed, we introduced the concept mainly because using bracketing numbers directly would have given slightly worse bounds—albeit the same order of growth—and somewhat obscured the proof of Theorem 19.

Lemma 14 (Variance bracketing complexity and L^2-bracketing numbers)**.** *Let Z be a random variable, $\mathcal{F} \subseteq L^2(P_Z)$, and $\delta > 0$. Denote L^2-bracketing numbers by*

$$N_Z(\varepsilon,\mathcal{F}) := N_{[]}\big(\varepsilon,\mathcal{F},L^2(P_Z)\big) \leq \infty,$$

i. e., $N_Z(\varepsilon,\mathcal{F})$ is the minimum number of brackets $[\ell,u]$ with $\ell,u \in L^2(P_Z)$ and $\|u-\ell\|_{L^2(P_Z)} < \varepsilon$ covering $\mathcal{F}$; set $N_Z(\varepsilon,\mathcal{F}) := \infty$ if no such cover exists. Consider

any $g_0 \in L^2(P_Z)$ *and any finite sequence* $(\varepsilon_k)_{k=0}^K$, $K \geq 1$, *with* $\varepsilon_0 > \varepsilon_1 > \ldots > \varepsilon_K > 0$ *such that* $\varepsilon_0 > \sup_{f \in \mathcal{F}} \sigma_Z(f, g_0)$ *and* $\varepsilon_K < \delta$. *Then,*

$$\mathcal{V}_Z(\delta, \mathcal{F}) \leq \sqrt{\operatorname{Var} g_0(Z)} + \sum_{k=1}^{K} \varepsilon_{k-1} \sqrt{N_Z(\varepsilon_k, \mathcal{F})} + \varepsilon_K \sqrt{N_Z(\varepsilon_K, \mathcal{F})}.$$

A proof is given in the appendix. The second part of our detour is an analogue to Lemma 3, where we observed that Rademacher complexity is invariant to constant shifts. Variance bracketing complexity has a similar property:

Lemma 15 (Invariance under constant shifts)**.** *Let* Z *be a random variable with values in some measurable space* $(S, \mathcal{S})$, *and let* $\mathcal{F}$ *and* $\mathcal{H}$ *be subsets of* $L^2(P_Z)$. *Suppose that every function in either set differs by a constant from some function in the other set. Let* $\delta > 0$.

(a) If $\sup_{h \in \mathcal{H}} \big|\mathrm{E} h(Z)\big| < \infty$, *then* $\mathcal{V}_Z(\delta, \mathcal{H}) \leq \mathcal{V}_Z(\delta, \mathcal{F})$.

(b) $\mathcal{V}_Z(\delta, \mathcal{H})$ *being finite implies* $\sup_{h \in \mathcal{H}} \big|\mathrm{E} h(Z)\big| < \infty$.

In particular, whenever both $\mathcal{V}_Z(\delta, \mathcal{F})$ *and* $\mathcal{V}_Z(\delta, \mathcal{H})$ *are finite,* $\mathcal{V}_Z(\delta, \mathcal{F}) = \mathcal{V}_Z(\delta, \mathcal{H})$.

This lemma, a proof of which can be found in the appendix, completes our detour. Next, we bound variance bracketing complexity of half-lines, for the uniform distribution on the unit interval:

Lemma 16 (Variance bracketing complexity of half-lines)**.** *Let*

$$\mathcal{I} := \big\{ \mathbf{1}_{(-\infty, t]} : t \in \mathbb{R} \big\}.$$

With respect to the uniform distribution on $[0, 1]$, *for every* $\delta > 0$,

$$\mathcal{V}_{\mathrm{U}[0,1]}(\delta, \mathcal{I}) \leq \left\lceil \frac{1}{2} \log_2 \frac{1}{\delta} \right\rceil^+.$$

A proof is given in the appendix. In order to apply the lemma, the following statement about transformation of real-valued into uniform random variables is useful:

Lemma 17 (Uniform transformation)**.** *Let* Z *be a real-valued random variable. Introduce an independent random variable* V, *uniformly distributed on* $[0, 1]$, *and define*

$$Z' := F_Z(Z-) + V\big(F_Z(Z) - F_Z(Z-)\big).$$

Then Z' *is uniformly distributed on* $[0, 1]$, *and, for every* $z \in \mathbb{R}$,

$$\mathbf{1}_{\{Z \leq z\}} = \mathbf{1}_{\{Z' \leq F_Z(z)\}} \quad a.\,s.$$

A proof is given in the appendix.

3.2. Influence result for the pairwise independent case

The following observation [1, Prop. 2.1], which is an application of Bessel's inequality to pairwise independent variables, is central to our argument:

Lemma 18 (Variance decomposition)**.** *Let $X_1, \dots, X_n$ be pairwise independent random variables, Z a random variable with values in a measurable space $(S, \mathcal{S})$, and $f \in L^2(P_Z)$. Then,*

$$\mathrm{E} \sum_{i=1}^{n} \Big(\mathrm{E}\big[f(Z) \bigm| X_i \big] - \mathrm{E} f(Z) \Big)^2 \le \operatorname{Var} f(Z).$$

Proof. Let

$$Y := f(Z) - \sum_{i=1}^{n} \mathrm{E}\big[f(Z) \bigm| X_i \big].$$

By the pairwise independence of the random variables X_i, the conditional expectations $\mathrm{E}\big[f(Z) \bigm| X_i \big]$ are pairwise uncorrelated. Therefore, for every j,

$$\begin{aligned} \operatorname{Cov}\Big(Y, \mathrm{E}\big[f(Z) \bigm| X_j \big] \Big) &= \operatorname{Cov}\Big(f(Z), \mathrm{E}\big[f(Z) \bigm| X_j \big] \Big) \\ &\qquad - \sum_{i=1}^{n} \operatorname{Cov}\Big(\mathrm{E}\big[f(Z) \bigm| X_i \big], \mathrm{E}\big[f(Z) \bigm| X_j \big] \Big) \\ &= \operatorname{Var} \mathrm{E}\big[f(Z) \bigm| X_j \big] - \operatorname{Var} \mathrm{E}\big[f(Z) \bigm| X_j \big] \\ &= 0. \end{aligned}$$

Since thus the conditional expectations $\mathrm{E}\big[f(Z) \bigm| X_i \big]$ as well as Y are pairwise uncorrelated, we obtain

$$\begin{aligned} \operatorname{Var} f(Z) &= \operatorname{Var}\Big(Y + \sum_{i=1}^{n} \mathrm{E}\big[f(Z) \bigm| X_i \big] \Big) \\ &= \operatorname{Var} Y + \sum_{i=1}^{n} \operatorname{Var} \mathrm{E}\big[f(Z) \bigm| X_i \big] \\ &\ge \sum_{i=1}^{n} \operatorname{Var} \mathrm{E}\big[f(Z) \bigm| X_i \big] \\ &= \mathrm{E} \sum_{i=1}^{n} \Big(\mathrm{E}\big[f(Z) \bigm| X_i \big] - \mathrm{E} f(Z) \Big)^2, \end{aligned}$$

which is the desired inequality. □

Theorem 19 (Pairwise independent case)**.** *Let $X_1, \dots, X_n$ be pairwise independent random variables, Z a random variable, and $\mathcal{F}$ a Z-admissible set. Then, for every $\delta > 0$,*

$$\mathrm{E} \sum_i \operatorname*{ess\,sup}_{f \in \mathcal{F}} \Big(\big| \mathrm{E}\big[f(Z) \bigm| X_i \big] - \mathrm{E} f(Z) \big| - \delta \Big)_+^2 \le \mathcal{V}_Z(\delta, \mathcal{F})^2. \tag{8}$$

In particular, for every $\delta > 0$,

$$\mathrm{E} \sum_i \operatorname*{ess\,sup}_{f \in \mathcal{F}} \Big(\mathrm{E}\big[f(Z) \bigm| X_i \big] - \mathrm{E} f(Z) \Big)^2 \le \Big(\mathcal{V}_Z(\delta, \mathcal{F}) + \delta \sqrt{n} \Big)^2. \tag{9}$$

Proof. Fix an admissible triple $\big(K, (\mathcal{W}_k)_{k=0}^K, \mathcal{C}\big)$ as in the definition of $\mathcal{V}_Z(\delta, \mathcal{F})$, and write $\mathcal{W}_0 = \{g_0\}$. Write

$$d_f(X_i) := \mathrm{E}\big[f(Z) \bigm| X_i \big] - \mathrm{E} f(Z) \quad \text{and} \quad \Delta_i(f, g) := d_{f-g}(X_i).$$

The proof has three parts: (I) a chaining argument controlling the conditional deviations of the witnesses in $\mathcal{W}_K$, leading to (12); (II) a bracketing argument extending this control to $\mathcal{F}$, leading to (14); and (III) the combination of these two bounds.

(I) Chaining argument. For each $k = 1, \dots, K$ and $w \in \mathcal{W}_k$, choose $p_k(w) \in \mathcal{W}_{k-1}$ such that

$$\sigma_Z\big(w, p_k(w)\big) = \sigma_Z(w, \mathcal{W}_{k-1}).$$

Such a choice is possible since $\mathcal{W}_{k-1}$ is nonempty and finite.

By variance decomposition (Lemma 18), we have

$$\begin{aligned} \sum_i \mathrm{E} \max_{w \in \mathcal{W}_k} \big| \Delta_i\big(w, p_k(w)\big) \big|^2 &\le \sum_{w \in \mathcal{W}_k} \sum_i \mathrm{E} \big| \Delta_i\big(w, p_k(w)\big) \big|^2 \\ &\le \sum_{w \in \mathcal{W}_k} \sigma_Z^2\big(w, p_k(w)\big) = \sum_{w \in \mathcal{W}_k} \sigma_Z^2(w, \mathcal{W}_{k-1}). \end{aligned} \tag{10}$$

For any $w \in \mathcal{W}_K$, define its chain recursively by $w_K := w$ and $w_{k-1} := p_k(w_k)$ for $k = K, \dots, 1$. Since $\mathcal{W}_0 = \{g_0\}$, we have $w_0 = g_0$. By telescoping,

$$d_w(X_i) = d_{g_0}(X_i) + \sum_{k=1}^K \Delta_i(w_k, w_{k-1}).$$

Consequently,

$$\max_{w \in \mathcal{W}_K} \big| d_w(X_i) \big| \le \big| d_{g_0}(X_i) \big| + \sum_{k=1}^K \max_{w \in \mathcal{W}_k} \big| \Delta_i\big(w, p_k(w)\big) \big|.$$

Minkowski's inequality gives

$$\left(\sum_i \mathrm{E} \max_{w\in\mathcal{W}_K} \bigl|d_w(X_i)\bigr|^2\right)^{1/2}$$
$$\le \left(\sum_i \mathrm{E}\bigl|d_{g_0}(X_i)\bigr|^2\right)^{1/2} + \sum_{k=1}^{K} \left(\sum_i \mathrm{E} \max_{w\in\mathcal{W}_k} \bigl|\Delta_i\bigl(w, p_k(w)\bigr)\bigr|^2\right)^{1/2}. \tag{11}$$

Together with (10), we obtain

$$\left(\sum_i \mathrm{E} \max_{w\in\mathcal{W}_K} |d_w(X_i)|^2\right)^{1/2} \le \sqrt{\operatorname{Var} g_0(Z)} + \sum_{k=1}^{K} \sqrt{\sum_{w\in\mathcal{W}_k} \sigma_Z^2(w, \mathcal{W}_{k-1})}. \tag{12}$$

(II) Bracketing argument. For every bracket $[\ell, u] \in \mathcal{C}$, choose $w_\ell, w_u \in \mathcal{W}_K$ such that

$$\sigma_Z(\ell, w_\ell) = \sigma_Z(\ell, \mathcal{W}_K),$$
$$\sigma_Z(u, w_u) = \sigma_Z(u, \mathcal{W}_K).$$

By variance decomposition (Lemma 18) and these choices,

$$\sum_i \mathrm{E} \max_{[\ell,u]\in\mathcal{C}} \max\bigl\{|\Delta_i(\ell, w_\ell)|^2,\ |\Delta_i(u, w_u)|^2\bigr\}$$
$$\le \sum_{[\ell,u]\in\mathcal{C}} \sum_i \mathrm{E}\Bigl[|\Delta_i(\ell, w_\ell)|^2 + |\Delta_i(u, w_u)|^2\Bigr]$$
$$\le \sum_{[\ell,u]\in\mathcal{C}} \bigl(\sigma_Z^2(\ell, w_\ell) + \sigma_Z^2(u, w_u)\bigr)$$
$$= \sum_{[\ell,u]\in\mathcal{C}} \bigl(\sigma_Z^2(\ell, \mathcal{W}_K) + \sigma_Z^2(u, \mathcal{W}_K)\bigr). \tag{13}$$

Now consider any $f \in \mathcal{F}$ and a bracket $[\ell, u] \in \mathcal{C}$ containing f. Let $\varepsilon := \mathrm{E}\bigl((u-\ell)(Z)\bigr)$. Since $\ell(Z) \le f(Z) \le u(Z)$ a. s., we have $\mathrm{E}\bigl[\ell(Z) \mid X_i\bigr] \le \mathrm{E}\bigl[f(Z) \mid X_i\bigr] \le \mathrm{E}\bigl[u(Z) \mid X_i\bigr]$ a. s. and also $-\mathrm{E}u(Z) \le -\mathrm{E}f(Z) \le -\mathrm{E}\ell(Z)$, which gives

$$d_\ell(X_i) - \varepsilon \le d_f(X_i) \le d_u(X_i) + \varepsilon$$

and thus

$$|d_f(X_i)| \le \max\bigl\{|d_\ell(X_i)|,\ |d_u(X_i)|\bigr\} + \varepsilon.$$

Using the definition of Δ_i and bounding by the maxima, we obtain

$$|d_f(X_i)| \le \max_{w\in\mathcal{W}_K} |d_w(X_i)|$$
$$+ \max_{[\ell,u]\in\mathcal{C}} \max\bigl\{|\Delta_i(\ell, w_\ell)|,\ |\Delta_i(u, w_u)|\bigr\}$$
$$+ \max_{[\ell,u]\in\mathcal{C}} \mathrm{E}\bigl[(u-\ell)(Z)\bigr].$$

Bounding the last term by δ and taking it to the left-hand side then gives

$$\bigl(|d_f(X_i)| - \delta\bigr)^+ \le \max_{w\in\mathcal{W}_K} |d_w(X_i)| + \max_{[\ell,u]\in\mathcal{C}} \max\bigl\{|\Delta_i(\ell, w_\ell)|,\ |\Delta_i(u, w_u)|\bigr\}.$$

Since the right-hand side is independent of f, it also bounds the essential supremum over $f \in \mathcal{F}$ a. s. Minkowski's inequality and (13) therefore give

$$\begin{aligned}\left(\sum_i \mathrm{E}\operatorname*{ess\,sup}_{f\in\mathcal{F}} \bigl(|d_f(X_i)| - \delta\bigr)_+^2\right)^{1/2} &\le \left(\sum_i \mathrm{E}\max_{w\in\mathcal{W}_K} |d_w(X_i)|^2\right)^{1/2} \\ &\quad + \sqrt{\sum_{[\ell,u]\in\mathcal{C}} \bigl(\sigma_Z^2(\ell, \mathcal{W}_K) + \sigma_Z^2(u, \mathcal{W}_K)\bigr)}. \qquad (14)\end{aligned}$$

(III) Combination. Combining (14) with (12), taking the infimum over all admissible triples and squaring both sides proves assertion (8) of the theorem. For assertion (9), observe that $|a| \le \bigl(|a| - \delta\bigr)^+ + \delta$ and apply Minkowski's inequality. □

Corollary 20 (Cumulative distribution functions, pairwise independent case)**.** *Let $X_1, \dots, X_n$ be pairwise independent random variables and Z a real-valued random variable. Then, for every $\delta > 0$,*

$$\sum_i \mathrm{E}\sup_{z\in\mathbb{R}} \Bigl(\bigl|F_Z\bigl(z \mid X_i\bigr) - F_Z(z)\bigr| - \delta\Bigr)_+^2 \le \left\lceil \frac{1}{2}\log_2\frac{1}{\delta}\right\rceil^2. \qquad (15)$$

In particular,

$$\sum_i \mathrm{E}\sup_{z\in\mathbb{R}} \Bigl(F_Z\bigl(z \mid X_i\bigr) - F_Z(z)\Bigr)^2 \le \left\lceil 1 + \frac{1}{4}\log_2 n\right\rceil^2. \qquad (16)$$

Proof. We first consider the case where Z is uniformly distributed on $[0, 1]$. The set $\mathcal{I}$ is Z-admissible. By right-continuity of the distribution functions, the supremum over thresholds can be restricted to rational thresholds and therefore agrees almost surely with the essential supremum. Now assertion (15) follows from inequality (8) in Theorem 19 and Lemma 16. The same lemma and inequality (9) in Theorem 19 with $\delta = n^{-1/2}$ give assertion (16).

For general Z, introduce a random variable V, uniformly distributed on $[0, 1]$ and independent of everything else, and define the transformation

$$Z' := F_Z(Z-) + V\bigl(F_Z(Z) - F_Z(Z-)\bigr)$$

as in Lemma 17. By the lemma, Z' is uniformly distributed on $[0, 1]$ and

$$\mathbf{1}_{(-\infty,s]}(Z) = \mathbf{1}_{(-\infty,F_Z(s)]}(Z') \text{ a. s.}$$

Hence,

$$\sup_{z\in\mathbb{R}} \Bigl|F_Z\bigl(z \mid X_i\bigr) - F_Z(z)\Bigr| \le \sup_{s\in[0,1]} \Bigl|F_{Z'}\bigl(s \mid X_i\bigr) - F_{Z'}(s)\Bigr| \quad \text{a. s.},$$

and the assertions follow from what we have proved for the uniform case. □

4. Some lower bounds

In this section, we show that the order established in Corollary 20 is sharp in general. More precisely, we demonstrate that, under mere pairwise independence of the X_i, the expected squared maximum deviation of the conditional cumulative distribution function does not admit the constant bound available in the fully independent case. To this end, we take the X_i to be symmetric Rademacher random variables. The following lemma provides a lower bound for the quantity of interest in this setting.

Lemma 21 (Lower bound, Rademacher signs family)**.** *Let $X_1, \dots, X_n$ be random variables with $P(X_i = 1) = P(X_i = -1) = \frac{1}{2}$, let Z be a random variable and $\mathcal{F} \subseteq L^2(P_Z)$ a Z-admissible family. We do not make any independence assumptions. Then,*

$$\mathrm{E} \sum_i \sup_{f \in \mathcal{F}} \Big(\mathrm{E}\big[f(Z) \mid X_i\big] - \mathrm{E} f(Z) \Big)^2 \geq \sup \left\{ \frac{1}{n} \left(\sum_i \mathrm{E}\big[X_i f_i(Z)\big] \right)^2 : f_1, \dots, f_n \in \mathcal{F} \right\}.$$

Proof. We have, for any $f \in \mathcal{F}$,

$$\begin{aligned}
&\mathrm{E}\big[f(Z) \mid X_i\big] - \mathrm{E} f(Z) \\
&\quad = \mathrm{E}\big[f(Z) \mid X_i\big] - \left(\frac{1}{2} \mathrm{E}\big[f(Z) \mid X_i = 1\big] + \frac{1}{2} \mathrm{E}\big[f(Z) \mid X_i = -1\big] \right) \\
&\quad = \frac{1}{2} X_i \Big(\mathrm{E}\big[f(Z) \mid X_i = 1\big] - \mathrm{E}\big[f(Z) \mid X_i = -1\big] \Big) \\
&\quad = X_i \mathrm{E}\big[X_i f(Z)\big].
\end{aligned}$$

In particular,

$$\Big(\mathrm{E}\big[f(Z) \mid X_i\big] - \mathrm{E} f(Z) \Big)^2 = \Big(\mathrm{E}\big[X_i f(Z)\big] \Big)^2$$

is deterministic, and

$$\mathrm{E} \sum_i \sup_{f \in \mathcal{F}} \Big(\mathrm{E}\big[f(Z) \mid X_i\big] - \mathrm{E} f(Z) \Big)^2 = \sum_i \sup_{f \in \mathcal{F}} \Big(\mathrm{E}\big[X_i f(Z)\big] \Big)^2. \tag{17}$$

For all $f_1, \dots, f_n \in \mathcal{F}$,

$$\begin{aligned}
\left| \sum_i \mathrm{E}\big[X_i f_i(Z)\big] \right| &\leq \sum_i \Big| \mathrm{E}\big[X_i f_i(Z)\big] \Big| \leq \sum_i \sup_{f \in \mathcal{F}} \Big| \mathrm{E}\big[X_i f(Z)\big] \Big| \\
&\leq \left[n \cdot \sum_i \sup_{f \in \mathcal{F}} \Big(\mathrm{E}\big[X_i f(Z)\big] \Big)^2 \right]^{1/2},
\end{aligned}$$

where the last inequality follows from Cauchy–Schwarz. Using this to bound the right-hand side of (17) from below, we reach the lemma. □

Defining Z. Let $X_1, \dots, X_n$ be symmetric Rademacher signs, and consider the partial-sum process

$$S_0 := 0, \quad S_1 := X_1, \quad S_2 := X_1 + X_2, \quad \dots, \quad S_n := X_1 + \cdots + X_n.$$

Let Z denote the smallest index in $\{0, \dots, n\}$ at which this process attains its maximum.

For $f_i = \mathbf{1}_{(-\infty; i-1]}$, we obtain

$$\begin{aligned}\sum_{i=1}^{n} \mathrm{E}\big[X_i f_i(Z)\big] = \sum_{i=1}^{n} \mathrm{E}\big[X_i\big(1 - \mathbf{1}\{Z \geq i\}\big)\big] &= \sum_{i=1}^{n} \mathrm{E}X_i - \sum_{i=1}^{n} \mathrm{E}\big[X_i \mathbf{1}\{Z \geq i\}\big] \\ &= -\mathrm{E}\sum_{i=1}^{Z} X_i = -\mathrm{E}\max_{0 \leq k \leq n} S_k.\end{aligned}$$

Consequently, Lemma 21 yields

$$\mathrm{E}\sum_{i} \sup_{z \in \mathbb{R}} \big(F_Z(z \mid X_i) - F_Z(z)\big)^2 \geq \frac{1}{n}\left[\mathrm{E}\max_{0 \leq k \leq n} S_k\right]^2. \tag{18}$$

Independent variables. For comparison with the pairwise independent case considered below, we first suppose that $X_1, \dots, X_n$ are independent, so that (S_k) is a simple random walk. By the reflection principle, for every integer $m \geq 1$,

$$P\left(\max_{0 \leq k \leq n} S_k \geq m\right) = P(S_n \geq m) + P(S_n \geq m+1),$$

since reflecting a path after its first visit to m maps the paths that reach m but end below m bijectively onto the paths that end above m. Summing over m, we obtain

$$\begin{aligned}\mathrm{E}\left[\max_{0 \leq k \leq n} S_k\right] &= \mathrm{E}\big[S_n^+\big] + \mathrm{E}\big[(S_n - 1)^+\big] \\ &= 2\mathrm{E}\big[S_n^+\big] - P(S_n \geq 1) \\ &\geq \mathrm{E}|S_n| - \frac{1}{2},\end{aligned}$$

where the second step uses that S_n is integer-valued, and the last step uses the symmetry of S_n, which gives $\mathrm{E}S_n^+ = \frac{1}{2}\mathrm{E}|S_n|$ and $P(S_n \geq 1) \leq \frac{1}{2}$. By Khintchine's inequality with Szarek's optimal constant [28, Theorem 1],

$$\mathrm{E}|S_n| \geq \sqrt{\frac{n}{2}}\,.$$

Since $\sqrt{\frac{n}{2}} - \frac{1}{2} \geq 0$, inequality (18) therefore yields

$$\mathrm{E}\sum_{i} \sup_{z \in \mathbb{R}} \big(F_Z(z \mid X_i) - F_Z(z)\big)^2 \geq \frac{1}{n} \cdot \left(\sqrt{\frac{n}{2}} - \frac{1}{2}\right)^2 = \frac{\big(\sqrt{2n} - 1\big)^2}{4n}\,.$$

This bound exceeds $1/4$ precisely when $\sqrt{n} > \sqrt{2}+1$, that is, for all $n \geq 6$, and it tends to $1/2$ as $n \to \infty$. Using the asymptotics $\mathrm{E}|S_n| \sim \sqrt{2n/\pi}$ instead of Szarek's bound, the limit improves to $2/\pi$.

For a single threshold, by contrast, Lemma 18 gives, for every $z \in \mathbb{R}$,

$$\mathrm{E}\sum_i \Big(F_Z(z \mid X_i) - F_Z(z)\Big)^2 \leq \operatorname{Var} \mathbf{1}_{Z \leq z} \leq \frac{1}{4}\,.$$

Thus, even for independent X_i, taking the supremum over z inside the sum genuinely increases the total influence—for $n \geq 6$, it exceeds the bound $1/4$ that holds at a single threshold.

Pairwise independent variables. We now turn to the more interesting case of pairwise independence. Narayanan [19, §1.1, Eq. (1), p. 63:2] constructs pairwise independent symmetric Rademacher signs $h_1, \ldots, h_n$ such that

$$\mathrm{E} \max_{0 \leq k \leq n} \left| \sum_{i=1}^{k} h_i \right| = \Omega\big(\sqrt{n} \log n\big).$$

To obtain a corresponding lower bound without the absolute value, observe that

$$\begin{aligned} \max\left\{ \mathrm{E} \max_{0 \leq k \leq n} \sum_{i=1}^{k} h_i,\ -\mathrm{E} \min_{0 \leq k \leq n} \sum_{i=1}^{k} h_i \right\} &\geq \frac{1}{2}\mathrm{E}\left(\max_{0 \leq k \leq n} \sum_{i=1}^{k} h_i - \min_{0 \leq k \leq n} \sum_{i=1}^{k} h_i \right) \\ &\geq \frac{1}{2}\mathrm{E} \max_{0 \leq k \leq n} \left| \sum_{i=1}^{k} h_i \right|. \end{aligned}$$

Thus, for each n, we can choose either $X_i := h_i$ for all $i = 1, \ldots, n$ or $X_i := -h_i$ for all $i = 1, \ldots, n$ to obtain pairwise independent symmetric Rademacher signs satisfying

$$\mathrm{E} \max_{0 \leq k \leq n} S_k = \Omega\big(\sqrt{n} \log n\big).$$

Combining this with (18), we obtain

$$\mathrm{E}\sum_i \sup_{z \in \mathbb{R}} \big(F_Z(z \mid X_i) - F_Z(z)\big)^2 = \Omega\big((\log n)^2\big). \tag{19}$$

Thus, the logarithmic-squared growth in Corollary 20 cannot, in general, be improved under pairwise independence.

The logarithmic loss is related to the classical Rademacher–Menshov maximal inequality. For an orthonormal system $Y_1, \ldots, Y_n$ and $S_k := \sum_{j=1}^{k} Y_j$, the classical Rademacher–Menshov maximal inequality [16, Lemma 1, p. 333] gives

$$\left\| \max_{k \leq n} |S_k| \right\|_2 \lesssim \sqrt{n}\, \log(n+1).$$

Narayanan's construction attains this scale for pairwise independent symmetric signs, already in the first moment. Inequality (18) then transfers that maximal-growth phenomenon to the aggregate squared influence of the conditional distribution functions, which we are interested in.

The proof of our upper bound for distribution functions followed the same multiscale principle as the standard dyadic proof of the Rademacher–Menshov inequality [16, proof of Lemma 1, pp. 333–335]. In the chaining argument of Theorem 19, Lemma 18 controlled the aggregate squared increments at each scale. For half-lines, this control is uniform over scales; adding the contributions of logarithmically many scales produces a logarithmic bound in L^2, and hence a squared logarithm in the aggregate squared influence.

Symmetrization. The joint distribution of the random vector $(Z, X_1, \dots, X_n)$ is not invariant under permutations of the X_i. At the cost of enlarging the value space, however, one can achieve exchangeability:

Lemma 22 (Symmetrization)**.** *Let $X_1, \dots, X_n$ be random variables taking values in $E_1, \dots, E_n$, respectively, and let Z be another random variable. Then there exist random variables $Y_1, \dots, Y_n$ taking values in the disjoint union $\bigsqcup_{i=1}^n E_i$ with the following properties:*

1. *For every permutation π of $\{1, \dots, n\}$, the distributions of $(Z, Y_1, \dots, Y_n)$ and $(Z, Y_{\pi(1)}, \dots, Y_{\pi(n)})$ coincide.*
2. *With $\bigoplus$ denoting the sum of measures on the disjoint union,*

$$\mathcal{L}(Y_i) = \frac{1}{n} \bigoplus_{j=1}^{n} \mathcal{L}(X_j).$$

3. *If $X_1, \dots, X_n$ are pairwise independent, then so are $Y_1, \dots, Y_n$.*
4. *For every Z-admissible family $\mathcal{F}$,*

$$\begin{aligned} &\mathrm{E} \sum_{i=1}^{n} \operatorname*{ess\,sup}_{f \in \mathcal{F}} \Big(\mathrm{E}\big[f(Z) \mid Y_i \big] - \mathrm{E} f(Z) \Big)^2 \\ &\qquad = \left(1 - \frac{1}{n} \right)^2 \mathrm{E} \sum_{i=1}^{n} \operatorname*{ess\,sup}_{f \in \mathcal{F}} \Big(\mathrm{E}\big[f(Z) \mid X_i \big] - \mathrm{E} f(Z) \Big)^2. \end{aligned}$$

The random variables Y_i are constructed[2] by tagging the X_i to preserve their relationship to Z and then uniformly randomly permuting them. To restore pairwise independence,

[2]The underlying construction of pairwise-independent random labels, obtained by mixing uniformly random permutations with constant vectors, already appears in Feller [8], Chapter V, Section 3, footnote 3, p. 127.

one randomly chooses between the shuffled vector and a vector of independent draws from the distribution of X_J, where J is random. The construction and a proof of the lemma are given in the appendix.

The application of Lemma 22 to the random variables constructed above together with the lower bound for the order of growth established in (19) gives the following corollary, with the statement about the uniform marginal distributions following from the construction detailed in the proof of the lemma.

Corollary 23 (Exact growth order under symmetry and pairwise independence)**.** *There exist pairwise independent random variables $Y_1, \ldots, Y_n$, each uniform on $2n$ points, and a random variable Z such that for every permutation π, $(Z, Y_1, \ldots, Y_n)$ and $(Z, Y_{\pi(1)}, \ldots, Y_{\pi(n)})$ are equal in distribution and*

$$\mathrm{E} \sup_{z \in \mathbb{R}} \Big(F_Z(z \mid Y_i) - F_Z(z) \Big)^2 = \Theta\left(\frac{(\log n)^2}{n} \right).$$

□

5. Extensions and examples

5.1. Pivotality

In analogy to the notion of (p, α)-pivotality as defined by Gradwohl et al. [11, p. 972] and following the weak-inequality convention used in [1, Def. 5.1], we define:

Definition 24 (Pivotality)**.** *Let $X_1, \ldots, X_n$ and Z be random variables and $\mathcal{F}$ a Z-admissible family. Let $0 < p \leq 1$ and $\alpha > 0$. An index i is called* (p, α)-pivotal for $(Z, \mathcal{F})$ *if*

$$P\left(\operatorname*{ess\,sup}_{f \in \mathcal{F}} \Big| \mathrm{E}\big(f(Z) \mid X_i\big) - \mathrm{E} f(Z) \Big| \geq \alpha \right) \geq p.$$

If Z is real-valued, the index i is called distributionally (p, α)-pivotal *for Z when*

$$P\left(\sup_{z \in \mathbb{R}} \Big| F_Z(z \mid X_i) - F_Z(z) \Big| \geq \alpha \right) \geq p.$$

With the results established above, we can now bound these quantities from above.

Proposition 25. *Let $X_1, \dots, X_n$ and Z be random variables and $\mathcal{F}$ a Z-admissible family. Let $0 < p \le 1$ and $\alpha > 0$. Then,*

$$\#\big\{i \in \{1,\dots,n\} : i \text{ is } (p,\alpha)\text{-pivotal for } (Z,\mathcal{F})\big\} \le \begin{cases} \dfrac{4\mathcal{R}_2(\mathcal{F})^2}{p\alpha^2} & \text{if } X_1,\dots,X_n \text{ are independent,} \\[2ex] \inf\limits_{\delta\in(0,\alpha)} \dfrac{\mathcal{V}_Z(\delta,\mathcal{F})^2}{p(\alpha-\delta)^2} & \text{if } X_1,\dots,X_n \text{ are pairwise independent.} \end{cases}$$

In particular, if Z is real-valued,

$$\#\big\{i \in \{1,\dots,n\} : i \text{ is distributionally } (p,\alpha)\text{-pivotal for } Z\big\} \le \begin{cases} \dfrac{4}{p\alpha^2} & \text{if } X_1,\dots,X_n \text{ are independent,} \\[2ex] \dfrac{4}{p\alpha^2}\cdot\left\lceil \dfrac{1}{2}\log_2\dfrac{2}{\alpha}\right\rceil^2 & \text{if } X_1,\dots,X_n \text{ are pairwise independent.} \end{cases}$$

Proof. Let $D_i := \operatorname{ess\,sup}_{f\in\mathcal{F}}\big|\mathrm{E}\big[f(Z) \;\big|\; X_i\big] - \mathrm{E}f(Z)\big|$ and I be the set of (p,α)-pivotal indices for $(Z,\mathcal{F})$. For every $i \in I$, and every $\delta \in [0,\alpha)$,

$$p(\alpha-\delta)^2 \le (\alpha-\delta)^2 P(D_i \ge \alpha) \le \mathrm{E}(D_i-\delta)_+^2.$$

Consequently,

$$p(\alpha-\delta)^2 \cdot \#I \le \sum_{i\in I}\mathrm{E}(D_i-\delta)_+^2 \le \mathrm{E}\sum_{i=1}^{n}\operatorname*{ess\,sup}_{f\in\mathcal{F}}\big(\big|\mathrm{E}[f(Z)\mid X_i] - \mathrm{E}f(Z)\big| - \delta\big)_+^2.$$

With $\delta = 0$, Theorem 7 gives the first bound for the independent case. Theorem 19, after taking the infimum over $\delta > 0$, the first bound for the pairwise independent case. For the distributional assertions, apply the same argument with $D_i := \sup_{z\in\mathbb{R}}\big|F_Z\big(z \mid X_i\big) - F_Z(z)\big|$ and use Corollary 9 with $\delta = 0$ and Corollary 20 with $\delta = \alpha/2$, respectively. □

We note that for $\mathcal{F} = \{\mathrm{id}\}$ and real-valued Z, (p,α)-pivotality for $(Z,\mathcal{F})$ reduces to the usual notion of (p,α)-pivotality. Taking $K = 0$, $\mathcal{W}_0 = \{\mathrm{id}\}$, and $\mathcal{C} = \{[\mathrm{id},\mathrm{id}]\}$ in Definition 13 gives $\mathcal{V}_Z\big(\delta,\{\mathrm{id}\}\big)^2 \le \operatorname{Var} Z$ for every $\delta > 0$; hence, letting $\delta \downarrow 0$, the proposition recovers the bound $(\operatorname{Var} Z)/(p\alpha^2)$ for the number of (p,α)-pivotal players obtained in Becker [1, Thm. 5.1].

5.2. Bounding by total variation

Definition 26 (Total variation, bounded variation)**.** *The* total variation *of a function* $v\colon \mathbb{R} \to \mathbb{R}$ *is*

$$\mathrm{TV}(v) := \sup\left\{ \sum_{j=1}^{m} \bigl|v(z_j) - v(z_{j-1})\bigr| : m \geq 1,\ z_0 < z_1 < \ldots < z_m,\ z_j \in \mathbb{R} \right\}.$$

We say that v *has* bounded variation *if* $\mathrm{TV}(v) < \infty$.

A standard result says that every function of bounded variation is bounded and Borel measurable.

Proposition 27. *Let* $X_1, \ldots, X_n$ *be random variables,* Z *a real-valued random variable, and let* $\mathcal{F}$ *be a* Z*-admissible family of functions* $f\colon \mathbb{R} \to \mathbb{R}$ *with* $\sup_{f\in\mathcal{F}} \mathrm{TV}(f) < \infty$*. Then,*

$$\mathrm{E} \sum_{i=1}^{n} \operatorname*{ess\,sup}_{f\in\mathcal{F}} \Bigl| \mathrm{E}\bigl[f(Z) \mid X_i\bigr] - \mathrm{E} f(Z) \Bigr|^2$$

$$\leq \Bigl(\sup_{f\in\mathcal{F}} \mathrm{TV}(f) \Bigr)^2 \cdot \begin{cases} 4 & \text{if } X_1, \ldots, X_n \text{ are independent,} \\ \left\lceil 1 + \frac{1}{4} \log_2 n \right\rceil^2 & \text{if } X_1, \ldots, X_n \text{ are pairwise independent.} \end{cases}$$

Proof. Assume $f\colon \mathbb{R} \to \mathbb{R}$ has bounded variation. Consider any cumulative distribution functions F and G, and let $D := \sup_{x\in\mathbb{R}} \bigl|F(x) - G(x)\bigr|$. Then, also $|F(x-) - G(x-)| \leq D$. Let g be any bounded nondecreasing function g. Then its superlevel sets are upper half-lines, possibly empty or all of $\mathbb{R}$, and hence their probabilities under F and G differ by at most D. Thus the layer-cake formula gives

$$\begin{aligned} \left| \int g\, dF - \int g\, dG \right| &\leq \int_{\inf g}^{\sup g} \Bigl| P_F\bigl(\{x : g(x) > t\}\bigr) - P_G\bigl(\{x : g(x) > t\}\bigr) \Bigr| \, \mathrm{d}t \\ &\leq (\sup g - \inf g) \cdot D. \end{aligned}$$

By Jordan decomposition, we can write $f = g - h$, where g, h are bounded nondecreasing functions satisfying

$$(\sup g - \inf g) + (\sup h - \inf h) = \mathrm{TV}(f).$$

Applying the preceding bound to g and h and using the triangle inequality yields

$$\left| \int_{\mathbb{R}} f(z)\, dF(z) - \int_{\mathbb{R}} f(z)\, dG(z) \right| \leq \mathrm{TV}(f) \cdot \sup_{z\in\mathbb{R}} \bigl|F(z) - G(z)\bigr|.$$

With $F(z) := F_Z(z \mid X_i)$ and $G(z) := F_Z(z)$, this becomes

$$\Big|\mathrm{E}\big[f(Z) \mid X_i\big] - \mathrm{E}f(Z)\Big| \le \mathrm{TV}(f) \cdot \sup_{z\in\mathbb{R}} \big|F_Z(z \mid X_i) - F_Z(z)\big| \quad \text{a. s.} \tag{20}$$

Squaring, taking the essential supremum, summing, taking the expected value and applying the Corollaries 9 and 20, respectively, yields the desired bounds. □

5.3. Two economic examples

Financing a public good with heterogeneous benefits. We illustrate our results with financing a nonexcludable public good. As in Mailath and Postlewaite [15] and Al-Najjar and Smorodinsky [17], incentive compatibility and voluntary participation limit contributions through agents' influence on provision. We move from binary provision and linear benefits to arbitrary quality rules and privately known, heterogeneous nonlinear benefit schedules. Similar utility structures appear in Xi and Xie [30, §3.4], who analyse dominant-strategy incentive-compatible and ex post individually rational mechanisms choosing between zero and full provision, and in Kunimoto and Zhang [14], whose Theorem 2 (p. 447) assumes that, as the population grows, changing an agent's report has a vanishing effect on the probability of provision (Condition α)—an assumption they justify for a restricted class of mechanisms in Proposition 2. In our model with positive-probability null types, the aggregate influence bound yields a financing restriction without imposing a nonpivotality assumption on the mechanism.

Consider agents $i = 1, \dots, n$ and a direct mechanism choosing a provision level $Q \in [0, 1]$ of a public good. Q can be thought of as the *quality* of the public good. Let $B > 0$ be a fixed, common upper bound on gross benefits. Each agent i has a privately known type θ_i, randomly and independently drawn from a finite set S_i. The agent's gross benefit from the public good is given by $v_i(\theta_i, Q)$, where we assume that for each type $s \in S_i$, the function $q \mapsto v_i(s, q)$ is non-negative and non-decreasing with values in $[0, B]$ and $v_i(s, 0) = 0$. We further assume there is a null type $0 \in S_i$ with $P(\theta_i = 0) = p_i \in (0, 1)$ and $v_i(0, q) = 0$ for all q.

The public good is financed through payments T_i of the agents. The utility of agent i is assumed to be given by

$$v_i(\theta_i, Q) - T_i.$$

Assume Bayesian incentive compatibility and interim individual rationality with outside utility zero; any randomization by the mechanism is assumed to be independent of the types.

Define

$$D_i(s) := \sup\left\{ \Big|\mathrm{E}\big[f(Q) \mid \theta_i = s\big] - \mathrm{E}f(Q)\Big| \;:\; f\colon \mathbb{R} \to [0, B] \text{ non-decreasing} \right\}.$$

Let

$$t_i(b) := \mathrm{E}[T_i \mid \theta_i = b]$$

be the expected payment of agent i, given his type is b. Since the types are independent, we can identify the outcome from reporting a certain type with the conditional outcome under that type. By the null type's incentive constraint against reporting s and by type s's constraint against reporting 0, we have

$$0 \le t_i(s) - t_i(0) \le \mathrm{E}\big[v_i(s,Q) \;\big|\; \theta_i = s\big] - \mathrm{E}\big[v_i(s,Q) \;\big|\; \theta_i = 0\big] \le D_i(s) + D_i(0).$$

This gives, by $(a+b)^2 \le a^2/p_i + b^2/(1-p_i)$,

$$\begin{aligned}
\mathrm{E}\big[\big(t_i(\theta_i) - t_i(0)\big)^2\big] &= \mathrm{E}\big[\mathbf{1}\{\theta_i \neq 0\} \cdot \big(t_i(\theta_i) - t_i(0)\big)^2\big] \\
&\le \frac{1}{p_i}\mathrm{E}\big[\mathbf{1}\{\theta_i \neq 0\} \cdot D_i(\theta_i)^2\big] + \frac{1}{1-p_i} \cdot P(\theta_i \neq 0) \cdot D_i(0)^2 \\
&= \frac{1}{p_i}\Big(\mathrm{E}\big[\mathbf{1}\{\theta_i \neq 0\} \cdot D_i(\theta_i)^2\big] + p_i D_i(0)^2\Big) \\
&= \frac{1}{p_i}\mathrm{E}D_i(\theta_i)^2.
\end{aligned}$$

Extending the benefit schedules $q \mapsto v_i(s,q)$ constantly to $\mathbb{R}$, we see that they belong to the class of functions over which the supremum defining D_i is taken. Every function in this class has total variation at most B, and Proposition 27 yields

$$\mathrm{E}\sum_i p_i\big(t_i(\theta_i) - t_i(0)\big)^2 \le 4B^2.$$

For expected revenue, the fact that $t_i(0) \le 0$ and Cauchy–Schwarz now give

$$\begin{aligned}
\mathrm{E}\sum_i T_i &\le \mathrm{E}\sum_i \mathbf{1}\{\theta_i \neq 0\}\big(t_i(\theta_i) - t_i(0)\big) \\
&\le \sqrt{\sum_i \frac{1-p_i}{p_i}} \cdot \sqrt{\mathrm{E}\sum_i p_i\big(t_i(\theta_i) - t_i(0)\big)^2} \\
&\le 2B\sqrt{\sum_i \frac{1-p_i}{p_i}}\,.
\end{aligned}$$

Assume the provision of the good costs $n \cdot c(Q)$, i. e. provision cost is proportional to the size of the population and determined by the quality via a strictly increasing function c with $c(0) = 0$. Ex-ante budget feasibility requires that expected revenue covers expected cost, which means $\mathrm{E}\sum_i T_i \ge \mathrm{E}nc(Q)$; our above bound therefore yields

$$\mathrm{E}c(Q) \le \frac{2B}{n} \cdot \sqrt{\sum_i \frac{1-p_i}{p_i}}\,,$$

and for every threshold $q \in (0, 1]$, we obtain

$$P(Q \geq q) \leq \frac{\mathrm{E}c(Q)}{c(q)} \leq \frac{2B}{nc(q)} \cdot \sqrt{\sum_i \frac{1-p_i}{p_i}}\,.$$

Hence, if

$$\sum_{i=1}^{n} \frac{1-p_i}{p_i} = \mathrm{o}(n^2) \text{ as } n \to \infty,$$

it follows that

$$Q \xrightarrow{P} 0 \text{ as } n \to \infty.$$

This condition is satisfied, in particular, if the null-type probabilities are uniformly bounded away from zero, i. e., if there exists $p_* > 0$ such that $p_i \geq p_*$ for all i and n. In this case, the probability of attaining a fixed positive provision level is $\mathrm{O}(n^{-1/2})$.

Since the revenue bound does not depend on the cost, the same argument shows, for null-type probabilities bounded away from zero, that $P(Q \geq q) \to 0$ whenever the cost of quality q grows faster than $\sqrt{n}$. The same square-root threshold appears in Hellwig's isoelastic model without crowding [12, §7] and in Xi and Xie [30]. By contrast, with a cost function independent of population size, Hellwig obtains convergence of second-best provision to first best in his bounded-provision setting [12, Proposition 3].

The standalone value of information. The value to a decision maker of observing a signal before acting—the increase in the maximal expected payoff obtained by conditioning on the signal—goes back to Blackwell [3] and Raiffa and Schlaifer [25, §4.5.2, pp. 89–91]. This setup is used widely, but typically with finitely many states: in the comparison of signals as complements or substitutes [4] and in the design and pricing of information [2], the decision problems have finite state and action spaces and the offered experiments have finite outcomes; Cabrales, Gossner, and Serrano [6], who identify actions with their state-contingent payoffs as we do below, allow infinitely many actions but finitely many states. In Bayesian evidence synthesis, Jackson et al. [13] formulate value of information for general decision problems with continuous parameters, developing finite-action and point-estimation cases to prioritize further research. In the following example, the state is arbitrary and the action set may be a continuum, at the price of a complexity assumption on the action set.

Suppose a decision maker chooses an action a from a nonempty, Z-admissible set $\mathcal{A}$ and receives the payoff $a(Z)$, where Z is random. Assume that

$$\sup_{a,z} \big|a(z)\big| < \infty \quad \text{and} \quad \mathcal{R}_2(\mathcal{A}) < \infty.$$

For example, a decision maker who estimates a real-valued Z by μ and is rewarded for accuracy could face $\mathcal{A} = \{a_\mu : \mu \in \mathbb{R}\}$ with $a_\mu(z) := \big(1 - |z-\mu|\big)^+$.[3]

[3] Writing $(1-|z-\mu|)^+ = \int_0^1 \big(\mathbf{1}\{z < \mu + r\} - \mathbf{1}\{z \leq \mu - r\}\big)\,\mathrm{d}r$ and moving the supremum over z in the definition of Rademacher complexity inside the integral shows $\mathcal{R}_2(\mathcal{A}) \leq 2\mathcal{R}_2(\mathcal{I})$, where $\mathcal{I}$ is the class of indicators of half-lines, which has finite Rademacher complexity by Lemma 4.

Suppose there are n potential signals $X_1, \dots, X_n$, independent of one another (unconditionally, not conditionally on Z), one of which can be observed before the action is chosen. The value of observing X_i is [25]

$$V_i := \mathrm{E} \operatorname*{ess\,sup}_{a \in \mathcal{A}} \mathrm{E}\big[a(Z) \;\big|\; X_i\big] - \sup_{a \in \mathcal{A}} \mathrm{E}a(Z),$$

which is nonnegative because $\operatorname{ess\,sup}_b \mathrm{E}[b(Z) \mid X_i] \geq \mathrm{E}[a(Z) \mid X_i]$ a. s. for every a. Let

$$S_i := \operatorname*{ess\,sup}_{a \in \mathcal{A}} \Big| \mathrm{E}\big[a(Z) \;\big|\; X_i\big] - \mathrm{E}a(Z) \Big|.$$

Since

$$\operatorname*{ess\,sup}_{a \in \mathcal{A}} \mathrm{E}[a(Z) \mid X_i] \leq \sup_{a \in \mathcal{A}} \mathrm{E}a(Z) + \operatorname*{ess\,sup}_{a \in \mathcal{A}} \big(\mathrm{E}[a(Z) \mid X_i] - \mathrm{E}a(Z) \big) \quad \text{a. s.},$$

we have $V_i \leq \mathrm{E}S_i$. By Jensen's inequality and Theorem 7,

$$\sum_{i=1}^n V_i^2 \leq \mathrm{E} \sum_{i=1}^n S_i^2 \leq 4\mathcal{R}_2(\mathcal{A})^2.$$

This means that the aggregate squared value of information is bounded[4] independently of the number of signals. In particular, at most $4\mathcal{R}_2(\mathcal{A})^2/\varepsilon^2$ signals can have value at least ε.

We should like to remark that the squares cannot be replaced by first powers: if n is odd, the X_i are independent Rademacher signs, $Z := \operatorname{sign}(X_1 + \cdots + X_n)$ and $\mathcal{A} = \big\{\mathrm{id}_{\{-1,1\}}, -\mathrm{id}_{\{-1,1\}}\big\}$, then

$$V_i = \binom{n-1}{\frac{1}{2}(n-1)} \cdot 2^{1-n} \sim \sqrt{\frac{2}{\pi n}} \quad \text{as } n \to \infty,$$

so $\sum_i V_i \to \infty$, while $\mathcal{R}_2(\mathcal{A})$ does not depend on n. For some decision problems with unbounded payoffs, however, such as estimation under quadratic loss [20] with finite prior variance or investment in complete markets with logarithmic utility [6, Lemma A.1, p.375], the value of a signal is an expected variance or entropy reduction, and the values themselves are summable.[5] Finally, our bound concerns each signal in isolation; independent signals can still be complements [4]. For example, if Z and X_1 are independent Rademacher signs and $X_2 := ZX_1$, then X_1 and X_2 are independent with $V_1 = V_2 = 0$, yet together they reveal $Z = X_1X_2$.

[4] If $\mathcal{A}$ is infinite, boundedness of the payoffs alone is not enough—in other words, the assumption $\mathcal{R}_2(\mathcal{A}) < \infty$ cannot be dropped. To see this, let Z be uniform on $[0,1)$, let X_i be its ith binary digit, and let $\mathcal{A} = \{\pm r_k : k \in \mathbb{N}^*\}$, where $r_k(z) = 1$ if the kth binary digit of z is 1 and $r_k(z) = -1$ otherwise. The payoffs are bounded by 1 and the signals are independent, yet $\sup_a \mathrm{E}a(Z) = 0$ and $V_i = 1$ for every i, so $\sum_{i=1}^n V_i^2 = n$. Accordingly, $\mathcal{R}_2(\mathcal{A}) = \infty$.

[5] For discrete Z and $B := \sup_{a,z} |a(z)|$, Pinsker's inequality [7, Lemma 11.6.1, p. 370], together with independence of the signals, yields $\sum_i V_i^2 \leq 2B^2 H(Z)$, where $H(Z)$ is the Shannon entropy in nats. This bound depends on the state entropy; differential entropy cannot replace $H(Z)$ for continuous Z.

Disclosure of AI use

This paper resolves a question that the author tried, unsuccessfully, to answer in 2017. The resolution became possible with the help of AI. The main arguments came from GPT-6 Astra and, to a lesser extent, Claude Opus 5.5. AI thus made *substantial contributions* to this paper.

When being provided with efforts on solving related problems as input and being asked for a bound on $\mathrm{E}\sum_i \sup_f \big(\mathrm{E}[f(Z) \mid X_i] - \mathrm{E}f(Z)\big)^2$, GPT-6 Astra produced essentially the proof of Theorem 7 and stated that independence of the X_i is necessary. When subsequently pushed towards a bracketing approach and being provided with some (better and worse) ideas, it produced a preliminary version of Theorem 19 and its proof. Both proofs were later refined, while retaining their essential arguments. GPT-6 Astra also found the construction of the random walk from Narayanan's [19] pairwise-independent Rademacher signs for establishing the lower bound. Claude Opus 5.5 refined the argument in the independent case. The initial ideas and main arguments for the economic examples in Section 5.3 likewise came from GPT-6 Astra. Both ChatGPT and Claude were also used for literature research, writing, and proofreading.

AI was used via chat on the respective web interface, the author selectively uploading relevant passages and material. The basic monthly subscription plans (ChatGPT Plus and Claude Team) were essentially sufficient.

The author takes full responsibility for the content of this paper.

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A. Proofs

Proof of Lemma 2. Set, for $f \in \mathcal{F}$,

$$B_f := \sum_{j:g_j=f} \epsilon_j c_j \,.$$

The random variables $(B_f)_{f\in\mathcal{F}}$ are independent and symmetric. Consequently, if $(\eta_f)_{f\in\mathcal{F}}$ is a family of independent Rademacher signs, independent of $(B_f)_{f\in\mathcal{F}}$, then

$$\big(B_f\big)_{f\in\mathcal{F}} \stackrel{d}{=} \big(\eta_f|B_f|\big)_{f\in\mathcal{F}}\,.$$

Hence,

$$\mathrm{E}_\epsilon \sup_{s\in S} \sum_{j=1}^m \epsilon_j c_j g_j(s) = \mathrm{E} \sup_{s\in S} \sum_{f\in\mathcal{F}} B_f f(s) = \mathrm{E}_B \mathrm{E}_\eta \sup_{s\in S} \sum_{f\in\mathcal{F}} \eta_f |B_f| f(s).$$

Since for every fixed realization (b_f) of B_f, we have, by the definition of $\mathcal{R}_2 := \mathcal{R}_2(\mathcal{F})$,

$$\mathrm{E}_\eta \sup_{s\in S} \sum_f \eta_f b_f f(s) \le \mathcal{R}_2 \cdot \bigg(\sum_{f\in\mathcal{F}} b_f^2\bigg)^{1/2},$$

it follows that

$$\mathrm{E}_\eta \sup_{s\in S} \sum_{f\in\mathcal{F}} \eta_f |B_f| f(s) \le \mathcal{R}_2 \cdot \bigg(\sum_{f\in\mathcal{F}} B_f^2\bigg)^{1/2}.$$

Therefore, by Jensen's inequality,

$$\mathrm{E}_\epsilon \sup_{s\in S} \sum_{j=1}^m \epsilon_j c_j g_j(s) \le \mathcal{R}_2 \cdot \mathrm{E}\bigg(\sum_{f\in\mathcal{F}} B_f^2\bigg)^{1/2} \le \mathcal{R}_2 \cdot \bigg(\mathrm{E}\sum_{f\in\mathcal{F}} B_f^2\bigg)^{1/2} = \mathcal{R}_2 \cdot \bigg(\sum_{j=1}^m c_j^2\bigg)^{1/2},$$

where the last equality follows from the independence and the centring of the Rademacher signs. □

Proof of Lemma 3. By symmetry, it suffices to show that $\mathcal{R}_2(\mathcal{F}) \le \mathcal{R}_2(\mathcal{H})$. If $\mathcal{F} = \emptyset$, this is immediate.

Let $f_1, \ldots, f_m \in \mathcal{F}$ be pairwise distinct. Choose $h_1, \ldots, h_m \in \mathcal{H}$ and $c_1, \ldots, c_m \in \mathbb{R}$ such that $h_j = f_j + c_j$ for every $j = 1, \ldots, m$.

For every $a = (a_1, \ldots, a_m) \in \mathbb{R}^m$ with $\|a\|_2 = 1$,

$$\mathrm{E}_\epsilon \sup_{s\in S} \sum_{j=1}^m \epsilon_j a_j f_j(s) = \mathrm{E}_\epsilon \sup_{s\in S} \sum_{j=1}^m \epsilon_j a_j h_j(s) \le \mathcal{R}_2(\mathcal{H}).$$

The equality holds because the two sums differ by $\sum_{j=1}^{m} \epsilon_j a_j c_j$, which is independent of s and has expectation zero. The inequality follows from Lemma 2, which applies even if the h_j are not distinct.

Taking the supremum over all such coefficient vectors a yields $\mathcal{R}_2(\{f_1, \ldots, f_m\}) \leq \mathcal{R}_2(\mathcal{H})$. Taking the supremum over all finite subsets of $\mathcal{F}$ proves the desired inequality. □

Proof of Lemma 4. Consider real thresholds $t_1 > \ldots > t_M$, independent Rademacher signs $\epsilon_1, \ldots, \epsilon_M$, and coefficients $a_1, \ldots, a_M$ with $\sum_i a_i^2 = 1$.

Let, for $0 \leq m \leq M$,

$$S_m := \sum_{i=1}^{m} \epsilon_i a_i \qquad \text{with} \qquad S_0 := 0.$$

Obviously,

$$\sup_{s \in \mathbb{R}} \sum_i \epsilon_i a_i \mathbf{1}_{s \leq t_i} = \max_{0 \leq m \leq M} S_m.$$

We are going to bound the expected value of this maximum.

For any $t > 0$, by Lévy's maximal inequality and the symmetry of S_M around 0,

$$P\left(\max_{0 \leq m \leq M} S_m \geq t \right) \leq 2P\big(S_M \geq t\big) = P\big(|S_M| \geq t\big).$$

Thus,

$$\begin{aligned} \mathrm{E} \max_{0 \leq m \leq M} S_m = \int_0^\infty P\left(\max_{0 \leq m \leq M} S_m \geq t \right) \mathrm{d}t \leq \int_0^\infty P\big(|S_M| \geq t\big) \mathrm{d}t = \mathrm{E}|S_M| \\ \leq \big(\mathrm{E} S_M^2\big)^{1/2} = \left(\sum_i a_i^2 \right)^{1/2} = 1. \end{aligned}$$

□

Proof of Lemma 15. For brevity, we write $Pg := \mathrm{E}g(Z)$.

(a) Consider any $\delta > 0$, and suppose $M := \sup_{h \in \mathcal{H}} |Ph| < \infty$.

Now take any admissible triple $\big(K, (\mathcal{W}_k)_{k=0}^K, \mathcal{C}\big)$ for $\mathcal{F}$. Since, by assumption,

$$\max_{[\ell, u] \in \mathcal{C}} P(u - \ell) < \delta,$$

some $\eta > 0$ exists such that

$$\max_{[\ell, u] \in \mathcal{C}} P(u - \ell) + \eta < \delta.$$

For each $[\ell, u] \in \mathcal{C}$, subdivide the interval

$$I_{\ell, u} := \big[-M - Pu, M - P\ell\big]$$

into finitely many intervals $[a,b]$ of length at most η. Then, form $\mathcal{C}'$ from all brackets

$$\big[\ell + a, u + b\big]$$

obtained this way. Then, every bracket $[\ell+a, u+b] \in \mathcal{C}'$ satisfies $P\big((u+b)-(\ell+a)\big) = P(u-\ell)+(b-a) \le P(u-\ell)+\eta < \delta$.

We are going to show that $\mathcal{C}'$ covers $\mathcal{H}$. Consider any $h \in \mathcal{H}$ and choose $f \in \mathcal{F}$ and $c \in \mathbb{R}$ such that $h = f + c$. Some $[\ell, u] \in \mathcal{C}$ exists containing f. Then,

$$c = Ph - Pf \in \big[Ph - Pu, Ph - P\ell\big] \subseteq \big[-M - Pu, M - P\ell\big] = I_{\ell,u}\,.$$

For a subdivision interval $[a,b]$ containing c,

$$\ell + a \le f + c \le u + b.$$

Hence, $[\ell+a, u+b] \in \mathcal{C}'$ covers $h = f + c$.

Defining an additional level by

$$\mathcal{W}_{K+1} := \big\{\ell, u : [\ell, u] \in \mathcal{C}\big\},$$

we thus obtain an admissible triple $\big(K+1, (\mathcal{W}_k)_{k=0}^{K+1}, \mathcal{C}'\big)$ for $\mathcal{H}$. Note that for any $[\ell+a, u+b] \in \mathcal{C}'$, since $\ell, u \in \mathcal{W}_{K+1}$,

$$\sigma_Z^2\big(\ell + a, \mathcal{W}_{K+1}\big) = \sigma_Z^2\big(\ell + a, \ell\big) = 0$$

and

$$\sigma_Z^2\big(u + b, \mathcal{W}_{K+1}\big) = \sigma_Z^2\big(u + b, u\big) = 0.$$

Further, since all elements of $\mathcal{W}_{K+1}$ are endpoints of the brackets in $\mathcal{C}$,

$$\sum_{w \in \mathcal{W}_{K+1}} \sigma_Z^2(w, \mathcal{W}_K) \le \sum_{[\ell,u]\in\mathcal{C}} \Big(\sigma_Z^2(\ell, \mathcal{W}_K) + \sigma_Z^2(u, \mathcal{W}_K)\Big).$$

Thus, with $\{g_0\} := \mathcal{W}_0$,

$$\begin{aligned}
\mathcal{V}_Z(\delta, \mathcal{H}) &\le \sqrt{\operatorname{Var} g_0(Z)} + \sum_{k=1}^{K+1} \sqrt{\sum_{w \in \mathcal{W}_k} \sigma_Z^2(w, \mathcal{W}_{k-1})} \\
&\le \sqrt{\operatorname{Var} g_0(Z)} + \sum_{k=1}^{K} \sqrt{\sum_{w \in \mathcal{W}_k} \sigma_Z^2(w, \mathcal{W}_{k-1})} \\
&\quad + \sqrt{\sum_{[\ell,u]\in\mathcal{C}} \Big(\sigma_Z^2(\ell, \mathcal{W}_K) + \sigma_Z^2(u, \mathcal{W}_K)\Big)}.
\end{aligned}$$

Taking the infimum over all admissible triples $\big(K, (\mathcal{W}_k)_{k=0}^{K}, \mathcal{C}\big)$ yields

$$\mathcal{V}_Z(\delta, \mathcal{H}) \le \mathcal{V}_Z(\delta, \mathcal{F}).$$

(b) Consider any $\delta > 0$ and assume that $\mathcal{V}_Z(\delta, \mathcal{H}) < \infty$. This implies that a non-empty finite bracket cover $\mathcal{C}$ of $\mathcal{H}$ exists, and it holds that

$$\min_{[\ell,u]\in\mathcal{C}} P\ell \le Ph \le \max_{[\ell,u]\in\mathcal{C}} Pu \quad \text{for all } h \in \mathcal{H}.$$

Hence,

$$\sup_{h\in\mathcal{H}} |Ph| \le \max\left\{ \Big|\min_{[\ell,u]\in\mathcal{C}} P\ell\Big|,\ \Big|\max_{[\ell,u]\in\mathcal{C}} Pu\Big| \right\} < \infty.$$

If both complexities are finite, part (b) gives bounded means for both families, and applying part (a) in both directions yields equality. □

Proof of Lemma 14. Suppose Z, $\mathcal{F}$, and $\delta > 0$ are given. Fix $g_0 \in L^2(P_Z)$, $R > \sup_{f\in\mathcal{F}} \sigma_Z(f, g_0)$, $K \ge 1$, and a family $(\varepsilon_k)_{k=0}^K$ with $\varepsilon_0 > \varepsilon_1 > \ldots > \varepsilon_K > 0$ such that $\varepsilon_0 \ge R$ and $\varepsilon_K < \delta$.

If $\mathcal{F} = \emptyset$, then $\mathcal{V}_Z(\delta, \mathcal{F}) = 0$, and the claim follows. Hence suppose $\mathcal{F} \neq \emptyset$. Suppose the bracketing numbers $N_Z(\varepsilon_k, \mathcal{F})$, $k = 0, \ldots, K$, are finite—otherwise there is nothing to prove. For each $k \ge 1$, choose a bracket cover $\mathcal{C}_k$ with exactly $N_Z(\varepsilon_k, \mathcal{F})$ brackets of L^2-width strictly smaller than ε_k. Since every bracket cover is of minimum size, we can, for each bracket $[\ell, u]$, select one representative $f_{[\ell,u]} \in [\ell, u] \cap \mathcal{F}$. Set

$$\mathcal{W}_0 := \{g_0\}, \qquad \mathcal{W}_k := \big\{ f_{[\ell,u]} : [\ell, u] \in \mathcal{C}_k \big\} \text{ for } k = 1, \ldots, K, \qquad \mathcal{C} := \mathcal{C}_K.$$

Since

$$\mathrm{E}\big[(u - \ell)(Z)\big] \le \|u - \ell\|_{L^2(P_Z)} < \varepsilon_K < \delta,$$

the triple $\big(K, (\mathcal{W}_k)_k, \mathcal{C}\big)$ is a triple admissible in Definition 13.

For $k \ge 2$, each $w \in \mathcal{W}_k \subseteq \mathcal{F}$ belongs to some bracket $[\ell, u] \in \mathcal{C}_{k-1}$. Since

$$\ell - u \le w - f_{[\ell,u]} \le u - \ell,$$

we have

$$|w - f_{[\ell,u]}| \le |u - \ell|$$

and hence

$$\sigma_Z(w, \mathcal{W}_{k-1}) \le \big\|w - f_{[\ell,u]}\big\|_{L^2(P_Z)} < \varepsilon_{k-1}\,.$$

For $k = 1$, the same bound follows from the definition of R and $\varepsilon_0 \ge R$. Thus, for $k = 1, \ldots, K$,

$$\sqrt{\sum_{w\in\mathcal{W}_k} \sigma_Z^2(w, \mathcal{W}_{k-1})} \le \varepsilon_{k-1}\sqrt{N_Z(\varepsilon_k, \mathcal{F})}.$$

At the terminal level K, the representative $f_{[\ell,u]}$ of each $[\ell, u] \in \mathcal{C}_K$ gives, since $\ell \le f_{[\ell,u]} \le u$,

$$\big(f_{[\ell,u]} - \ell\big)^2 + \big(u - f_{[\ell,u]}\big)^2 \le (u - \ell)^2.$$

Thus,

$$\sigma_Z^2(\ell, \mathcal{W}_K) + \sigma_Z^2(u, \mathcal{W}_K) \le \left\|f_{[\ell,u]} - \ell\right\|_{L^2(P_Z)}^2 + \left\|u - f_{[\ell,u]}\right\|_{L^2(P_Z)}^2$$
$$\le \|u - \ell\|_{L^2(P_Z)}^2 < \varepsilon_K^2.$$

Summing gives

$$\sqrt{\sum_{[\ell,u]\in\mathcal{C}} \left(\sigma_Z^2(\ell, \mathcal{W}_K) + \sigma_Z^2(u, \mathcal{W}_K)\right)} \le \varepsilon_K \sqrt{N_Z(\varepsilon_K, \mathcal{F})}.$$

Substituting the estimates into Definition 13 yields the claimed inequality. □

Proof of Lemma 16. Let U be uniformly distributed on $[0, 1]$.

If $\delta > 1$, we can use $K := 0$, $g_0 := 0$, and $\mathcal{C}$ consisting of the single bracket $[0, 1]$ to establish that $\mathcal{V}_U(\delta, \mathcal{I}) = 0$.

Now consider $\delta \in (0, 1]$. Let

$$K := \left\lceil \frac{1}{2} \log_2 \frac{1}{\delta} \right\rceil$$

and $g_0 := 0$.

Consider an arbitrary constant $c \in \left(1, 1 + 4^{-K}\right)$.

If $\delta = 1$, then $K = 0$; define

$$\mathcal{C} := \left\{ \left[0\,, \mathbf{1}_{(-\infty,1/c]}\right], \left[\mathbf{1}_{(-\infty,1/c]}\,, 1\right] \right\}.$$

Then, for $[\ell, u] \in \mathcal{C}$, we have

$$\mathrm{E}\big[(u - \ell)(U)\big] \le \max\left\{ \frac{1}{c}\,, 1 - \frac{1}{c} \right\} < 1 = \delta.$$

The definition of variance bracketing complexity gives

$$\mathcal{V}_U(1, \mathcal{I}) \le \sqrt{2 \operatorname{Var} \mathbf{1}_{(-\infty,1/c]}(U)} = \sqrt{\frac{2}{c}\left(1 - \frac{1}{c}\right)} = \frac{\sqrt{2(c-1)}}{c}\,.$$

Letting $c \downarrow 1$ proves the lemma for $\delta = 1$.

Now assume $\delta \in (0, 1)$. Take

$$\mathcal{W}_0 := \left\{\mathbf{1}_{(-\infty,0]}\right\}$$

and, for $k = 1, \ldots, K$,

$$\mathcal{W}_k := \left\{\mathbf{1}_{(-\infty,\, j/(c4^k)]} : j = 0, \ldots, 4^k\right\}.$$

Further, take the bracket cover $\mathcal{C}$ consisting of the brackets

$$\Big[\mathbf{1}_{(-\infty,\,(j-1)/(c4^K)]}\,,\,\mathbf{1}_{(-\infty,\,j/(c4^K)]}\Big],\quad j=1,\dots,4^K,$$

whose bracket widths are indeed

$$\mathrm{E}\Big[\big(\mathbf{1}_{(-\infty,\,j/(c4^K)]}-\mathbf{1}_{(-\infty,\,(j-1)/(c4^K)]}\big)(U)\Big]=\frac{1}{c4^K}\le\frac{1}{c4^{(1/2)\log_2(1/\delta)}}<\delta,$$

as well as the bracket

$$\Big[\mathbf{1}_{(-\infty,\,1/c]},1\Big],$$

whose bracket width is $1-1/c<1/(c4^K)<\delta$ as well.

We observe that, for $s,t\in[0,1]$,

$$\begin{aligned}\sigma_U^2\big(\mathbf{1}_{(-\infty,s]}\,,\,\mathbf{1}_{(-\infty,t]}\big)&=\mathrm{Var}\Big(\mathbf{1}\big\{\min\{s,t\}\le U\le\max\{s,t\}\big\}\Big)\\&=|s-t|\big(1-|s-t|\big).\end{aligned}\tag{21}$$

Now, for each $[\ell,u]\in\mathcal{C}$,

$$\sigma_U\big(\ell,\mathcal{W}_K\big)=0\quad\text{and}\,\sigma_U\big(u,\mathcal{W}_K\big)=0;$$

this follows from $\ell\in\mathcal{W}_K$ for all lower endpoints ℓ and from $u\in\mathcal{W}_K$ for all upper endpoints u except for the endpoint $u=1$ of the very last bracket, for which we have

$$\sigma_U\big(1,\mathcal{W}_K\big)\le\sigma_U(1,g_0)=\sigma_U(1,0)=0.$$

Hence, from the definition of variance bracketing complexity, we obtain

$$\mathcal{V}_U(\delta,\mathcal{I})\le\sum_{k=1}^{K}\sqrt{\sum_{w\in\mathcal{W}_k}\sigma_U^2(w,\mathcal{W}_{k-1})}.\tag{22}$$

First, we have

$$\sum_{w\in\mathcal{W}_1}\sigma_U^2(w,\mathcal{W}_0)=\sum_{w\in\mathcal{W}_1}\sigma_U^2(w,0)=\sum_{j=1}^{4}\frac{j}{4c}\left(1-\frac{j}{4c}\right)\le4\cdot\frac{1}{4}=1.\tag{23}$$

Now consider any $k=2,\dots,K$. Of the 4^k+1 points $w\in\mathcal{W}_k$, $4^{k-1}+1$ coincide with points in $\mathcal{W}_{k-1}$, and for them

$$\sigma_U^2\big(w,\mathcal{W}_{k-1}\big)=0.$$

Another 4^{k-1} points lie exactly in the middle of a coarse interval—for them, by (21),

$$\sigma_U^2\big(w,\mathcal{W}_{k-1}\big)\le\frac{2}{c4^k}\left(1-\frac{2}{c4^k}\right).$$

Finally, $2\cdot 4^{k-1}$ points lie exactly at one quarter or three quarters of a coarse interval—for them, by (21) and by considering the distance to the closer endpoint,

$$\sigma_U^2\big(w, \mathcal{W}_{k-1}\big) \le \frac{1}{c4^k}\left(1 - \frac{1}{c4^k}\right).$$

Hence,

$$\begin{aligned}\sum_{w\in\mathcal{W}_k} \sigma_U^2(w, \mathcal{W}_{k-1}) &\le (4^{k-1}+1)\cdot 0 + 4^{k-1}\cdot\frac{2}{c4^k}\left(1-\frac{2}{c4^k}\right) + 2\cdot 4^{k-1}\cdot\frac{1}{c4^k}\left(1-\frac{1}{c4^k}\right)\\ &= \frac{1}{2c}\left(2 - \frac{3}{c4^k}\right) = \frac{1}{c} - \frac{3}{2c^2\cdot 4^k} < \frac{1}{c}.\end{aligned}$$

Together with (22) and (23) we obtain $\mathcal{V}_U(\delta, \mathcal{I}) \le 1 + (K-1)/\sqrt{c} \le K$. This proves the lemma. □

Proof of Lemma 17. Fix $s \in (0,1)$, and set

$$z := \inf\big\{\zeta \in \mathbb{R} : F_Z(\zeta) > s\big\}.$$

Then z is finite, and $F_Z(z-) \le s \le F_Z(z)$.

If $Z < z$, then $Z' \le F_Z(Z) \le s$. If $Z > z$, there exists $y \in (z, Z)$ with $F_Z(y) > s$, so $Z' \ge F_Z(Z-) \ge F_Z(y) > s$. Therefore,

$$P(Z' \le s) = P(Z < z) + P(Z = z,\ Z' \le s).$$

If $F_Z(z-) < F_Z(z)$, the independence of V and Z gives

$$\begin{aligned}P(Z = z,\ Z' \le s) &= P\big(Z = z,\ F_Z(z-) + V(F_Z(z) - F_Z(z-)) \le s\big)\\ &= P\left(Z = z\right)\cdot P\left(V \le \frac{s - F_Z(z-)}{F_Z(z) - F_Z(z-)}\right)\\ &= \big(F_Z(z) - F_Z(z-)\big)\frac{s - F_Z(z-)}{F_Z(z) - F_Z(z-)}\\ &= s - F_Z(z-).\end{aligned}$$

The same identity holds when $F_Z(z-) = F_Z(z)$, since then $F_Z(z-) = F_Z(z) = s$ and $P(Z = z) = 0$. Consequently,

$$P(Z' \le s) = F_Z(z-) + s - F_Z(z-) = s.$$

Since this holds for every $s \in (0,1)$, Z' is uniformly distributed on $[0,1]$.

Consider any $z \in \mathbb{R}$. Then, $Z \le z$ implies $F_Z(Z) \le F_Z(z)$ and thus $Z' \le F_Z(z)$. Therefore, $\{Z \le z\} \subseteq \big\{Z' \le F_Z(z)\big\}$. As Z' is uniform on $[0,1]$,

$$P\Big(\big\{Z' \le F_Z(z)\big\} \setminus \{Z \le z\}\Big) = P\big(Z' \le F_Z(z)\big) - P(Z \le z) = F_Z(z) - F_Z(z) = 0.$$

□

Proof of Lemma 22. As in the proof of Theorem 7, we can assume that the variables X_i take values in pairwise disjoint measurable spaces, tagging them if necessary.

Introduce the following random variables, each of them independent from everything else:

- Π, which is uniform over all permutations of $\{1, \dots, n\}$;
- J, which is uniform over $\{1, \dots, n\}$;
- for each $i \in \{1, \dots, n\}$ and each $j \in \{1, \dots, n\}$, $\tilde{X}_i^{[j]}$ such that $\mathcal{L}\big(\tilde{X}_j^{[i]}\big) = \mathcal{L}(X_j)$;
- D, which takes the value 1 with probability $1/n$ and 0 with probability $1 - 1/n$.

For $i = 1, \dots, n$, define Y_i by

$$Y_i := \begin{cases} X_{\Pi(i)} & \text{if } D = 0, \\ \tilde{X}_J^{[i]} & \text{if } D = 1. \end{cases}$$

Statement 1 of the lemma follows directly from the construction. For statement 2, it suffices to observe that, conditional on either $\{D = 0\}$ or $\{D = 1\}$, the distribution of each Y_i is the uniform mixture of the distributions of $X_1, \dots, X_n$. To establish pairwise independence—statement 3—assume that $X_1, \dots, X_n$ are pairwise independent; then observe that, for $i \neq j$,

$$\begin{aligned}
\mathcal{L}\big((Y_i, Y_j)\big) &= \big(1 - \frac{1}{n}\big) \cdot \frac{1}{n!} \sum_{\pi \text{ permut.}} \mathcal{L}\big((X_{\pi(i)}, X_{\pi(j)})\big) + \frac{1}{n} \cdot \frac{1}{n} \sum_{k=1}^{n} \mathcal{L}(X_k) \otimes \mathcal{L}(X_k) \\
&= \big(1 - \frac{1}{n}\big) \cdot \frac{1}{n(n-1)} \sum_{(k,\ell),\, k \neq \ell} \mathcal{L}\big((X_k, X_\ell)\big) + \frac{1}{n^2} \sum_{k=1}^{n} \mathcal{L}(X_k) \otimes \mathcal{L}(X_k) \\
&= \frac{1}{n^2} \sum_{(k,\ell),\, k \neq \ell} \mathcal{L}(X_k) \otimes \mathcal{L}(X_\ell) + \frac{1}{n^2} \sum_{k=1}^{n} \mathcal{L}(X_k) \otimes \mathcal{L}(X_k) \\
&= \frac{1}{n^2} \sum_{(k,\ell)} \mathcal{L}(X_k) \otimes \mathcal{L}(X_\ell) \\
&= \left(\frac{1}{n} \sum_k \mathcal{L}(X_k)\right) \otimes \left(\frac{1}{n} \sum_\ell \mathcal{L}(X_\ell)\right) = \mathcal{L}(Y_i) \otimes \mathcal{L}(Y_j).
\end{aligned}$$

We now prove statement 4. First, assume $\mathcal{F}$ is finite. As in the proof of Theorem 7, for every $f \in \mathcal{F}$ choose a measurable function $x \mapsto P[x]f$ such that

$$P[X_i]f = \mathrm{E}\big[f(Z) \;\big|\; X_i\big] - \mathrm{E}f(Z) \quad \text{a. s. for every } i.$$

Then, using the defining property of the expected value, one shows that

$$\mathrm{E}\big[f(Z) \;\big|\; Y_i, D\big] - \mathrm{E}f(Z) = \mathbf{1}_{D=0} P[Y_i]f.$$

Since $P[Y_i]f$ is Y_i-measurable, conditioning on Y_i again gives

$$\mathrm{E}\big[f(Z) \mid Y_i\big] - \mathrm{E}f(Z) = \left(1 - \frac{1}{n}\right) P[Y_i]f.$$

Thus,

$$\mathrm{E}\sum_i \sup_{f\in\mathcal{F}} \Big(\mathrm{E}\big[f(Z) \mid Y_i\big] - \mathrm{E}f(Z)\Big)^2 = \left(1 - \frac{1}{n}\right)^2 \mathrm{E}\sum_{i=1}^n \sup_{f\in\mathcal{F}} \big(P[Y_i]f\big)^2.$$

Now, on the one hand,

$$\mathrm{E}\left[\mathbf{1}_{D=0} \sum_{i=1}^n \sup_{f\in\mathcal{F}} \big(P[Y_i]f\big)^2\right] = \mathrm{E}\left[\mathbf{1}_{D=0} \sum_{i=1}^n \sup_{f\in\mathcal{F}} \big(P[X_i]f\big)^2\right],$$

and on the other hand,

$$\begin{aligned}\mathrm{E}\left[\mathbf{1}_{D=1} \sum_{i=1}^n \sup_{f\in\mathcal{F}} \big(P[Y_i]f\big)^2\right] &= \mathrm{E}\left[\mathbf{1}_{D=1} \sum_{i=1}^n \sum_{j=1}^n P(J=j) \mathrm{E} \sup_{f\in\mathcal{F}} \big(P[\tilde{X}_j^i]f\big)^2\right] \\ &= \mathrm{E}\left[\mathbf{1}_{D=1} \sum_{j=1}^n \mathrm{E} \sup_{f\in\mathcal{F}} \big(P[X_j]f\big)^2\right] = \mathrm{E}\left[\mathbf{1}_{D=1} \sum_{j=1}^n \sup_{f\in\mathcal{F}} \big(P[X_j]f\big)^2\right].\end{aligned}$$

Therefore,

$$\mathrm{E}\sum_{i=1}^n \sup_{f\in\mathcal{F}} \Big(\mathrm{E}\big[f(Z) \mid Y_i\big] - \mathrm{E}f(Z)\Big)^2 = \left(1 - \frac{1}{n}\right)^2 \mathrm{E}\sum_{i=1}^n \sup_{f\in\mathcal{F}} \Big(\mathrm{E}\big[f(Z) \mid X_i\big] - \mathrm{E}f(Z)\Big)^2.$$

It remains to extend statement 4 to general Z-admissible families. Let $\mathcal{F}_0 = \{f_1, f_2, \ldots\} \subseteq \mathcal{F}$ be dense in $\mathcal{F}$ with respect to the $L^2(P_Z)$-seminorm, and set $\mathcal{F}_m := \{f_1, \ldots, f_m\}$. For any random variable W, define

$$d_f(W) := \mathrm{E}[f(Z) \mid W] - \mathrm{E}f(Z).$$

By the L^2-contraction property of conditional expectation,

$$\big\|d_f(W) - d_g(W)\big\|_{L^2(P)} \le \|f - g\|_{L^2(P_Z)}.$$

Consequently, the same density argument as in the proof of Theorem 7 yields

$$\operatorname*{ess\,sup}_{f\in\mathcal{F}} d_f(W)^2 = \sup_{m\ge 1} \max_{f\in\mathcal{F}_m} d_f(W)^2 \quad \text{a. s.}$$

Applying this with $W = X_i$ and $W = Y_i$, and using monotone convergence, we may pass to the limit $m \to \infty$ in the identity already established for the finite families $\mathcal{F}_m$, which completes the proof of statement 4. □